# THE TORELLI THEOREM FOR K3 SURFACES

YIWEN CAO

Abstract. The classical Torelli theorem states that the Jacobian of a curve, with its principal polarization, determines the curve. The phrase 'Torelli-type theorem' refers to results of this form. In this article, we focus on the case of K3 surfaces, which are a special class of complex surfaces. The Torelli theorem for K3 surfaces states that a lattice isomorphism of the second cohomology groups of two K3 surfaces that preserves additional data is induced by a unique isomorphism of complex surfaces. We give a proof of the result assuming that K3 surfaces are Kähler. The proof uses an interesting 'limit argument' and combines tools from algebraic geometry, Hodge theory, and complex geometry. Finally, we also include a proof of the surjectivity of the period map.

Contents

1. Introduction 1
2. Basics of K3 surfaces 3
3. The second cohomology groups of K3 surfaces 5
4. Complex tori and Kummer surfaces 9
5. Proof of the Torelli theorem for K3 surfaces 16
6. Surjectivity of the period map 26
Acknowledgments 28
References 29

## 1. Introduction

Given two complex manifolds, it is a fundamental problem to determine whether they are isomorphic, or biholomorphic. Even if they are diffeomorphic, this can be a very subtle problem. There are some invariants from algebraic topology which can distinguish complex manifolds. We will soon see that such (cohomological) invariants can sometimes even determine the underlying complex structure. When this happens, it is called a 'Torelli-type theorem'.

We first briefly recall the classical Torelli theorem. Let $C$ be a non-singular projective complex curve. For $[\gamma] \in H_1(C;\mathbb{Z})$, consider the map:

$$f_\gamma : H^0(C,\Omega^1_C) \to \mathbb{C}, \quad \omega \mapsto \int_\gamma \omega.$$

This gives a well-defined map

$$H_1(C,\mathbb{Z}) \to H^0(C,\Omega^1_C)^*,$$
$$[\gamma] \mapsto f_\gamma.$$

The map is injective because the intersection pairing is non-degenerate by Poincaré duality.

*Date*: September 19, 2026.

**Definition 1.1.** The *Jacobian* $J(C)$ of $C$ is defined to be the quotient $H^0(C,\Omega^1_C)^*/H_1(C;\mathbb{Z})$ by the above injection.

It comes with a principal polarization $\lambda_C$ given by the intersection pairing on $H_1(C;\mathbb{Z})$. The Torelli theorem states that the Jacobian with its principal polarization determines the curve.

**Theorem 1.2** (Torelli Theorem)**.** *If there is an isomorphism of principally polarized abelian varieties* $(J(C),\lambda_c)\cong(J(C'),\lambda_{C'})$*, then there is an isomorphism of curves* $C\cong C'$*.*

Notice that for an elliptic curve, the Jacobian is itself. Every elliptic curve can be expressed as $C=\mathbb{C}/\Lambda$ where $\Lambda$ is a free abelian group of rank 2. We have $H_1(C;\mathbb{Z})\cong\Lambda$. On the other hand, there is an isomorphism $\mathbb{C}^\vee\cong H^0(C,\Omega^1_C)$. This is because every holomorphic 1-form $\omega$ on $C$ pulls back to a holomorphic form $\pi^*\omega$ on $\mathbb{C}$. However, $\pi^*\omega$ is doubly-periodic with respect to $\Lambda$. Therefore the form $\pi^*\omega=f(z)\,dz$ has $f(z)=\lambda$ where $\lambda\in\mathbb{C}$. Hence, $\mathbb{C}^\vee\cong H^0(C,\Omega^1_C)$, and $\mathbb{C}\cong H^0(C,\Omega^1_C)^*$. We have

$$C=\mathbb{C}/\Lambda\cong H^0(C,\Omega^1_C)^*/H_1(C;\mathbb{Z})\cong J(C)$$

So the Torelli theorem is automatic for elliptic curves.

A Torelli-type theorem, or simply a Torelli theorem, refers to a generalization of this result to other kinds of varieties or complex manifolds.

This article deals with the family of K3 surfaces (see definition 2.1). They are a class of minimal complex surfaces that have Kodaira dimension 0. They are slightly more mysterious than surfaces that we understand well (e.g. rational surfaces or abelian surfaces) but more tractable to work with than, for example, surfaces of general type. The main result of this article, the Torelli theorem for K3 surfaces, is formulated as follows. For a K3 surface $X$, denote the non-vanishing holomorphic 2-form of $X$ by $\omega_X$ (see remark 2.2). Denote the Kähler cone of $X$ by $\mathcal{D}(X)$ (see definition 3.18).

**Theorem 1.3** (Torelli Theorem for K3 Surfaces)**.** *Let* $X,X'$ *be two K3 surfaces. Let*

$$\phi:H^2(X;\mathbb{Z})\longrightarrow H^2(X';\mathbb{Z})$$

*be a lattice isomorphism. Let* $\phi_{\mathbb{C}}:=\phi\otimes\mathbb{C}$*. If* $\phi_{\mathbb{C}}$ *satisfies*

*(1)* $\phi_{\mathbb{C}}(\omega_X)\in\mathbb{C}\omega_{X'}$*,*
*(2)* $\phi_{\mathbb{C}}(\mathcal{D}(X))=\mathcal{D}(X')$*.*

*Then there exists a unique isomorphism of complex surfaces* $\varphi:X'\to X$ *such that* $\varphi^*=\phi$*.*

It is hard to get our hands on this result at first sight. There is no 'canonical' model of a K3 surface. If there is such an isomorphism of cohomology groups between a quartic surface and a Kummer surface (see example 2.3 and definition 2.5), how can one find an actual isomorphism of complex surfaces?

The idea is to use a 'limit argument'. First we will prove the result for some special K3 surfaces. Then we will build a 'parameter space' of K3 surfaces in which we may approach any given K3 surface by a sequence of special K3 surfaces. Finally we show that the limit of the graph of the isomorphisms of such special K3 surfaces gives an actual isomorphism of the given K3 surfaces. The special K3 surfaces we choose are the Kummer surfaces. They are built up from complex tori, and their Torelli theorems can be proved directly.

The following is an outline of the article. In section 2 we give a basic introduction to K3 surfaces and discuss their basic properties. In section 3 we focus on the lattice structure of the second cohomology groups of K3 surfaces with some general lattice theory. In section 4 we discuss complex tori and Kummer surfaces and prove their Torelli theorems. In section 5 we state some results from complex deformation theory and prove the Torelli theorem for K3 surface. In section 6 we prove the surjectivity of the period map and as a corollary prove that not every K3 surface is projective.

The main reference for this article is [Kon20]. For basic results in algebraic topology or algebraic geometry, the reference is [Hat22] and [Har77], respectively. We also refer the readers to [LP80].

## 2. Basics of K3 surfaces

**Definition 2.1.** A compact connected complex surface $X$ is called a *K3 surface* if it has trivial canonical bundle and irregularity $q = 0$.

**Remark 2.2.** The condition that the canonical bundle is trivial is equivalent to the existence of a non-vanishing holomorphic 2-form (because the canonical bundle is a line bundle and a line bundle is trivial if and only if there is a nowhere zero section.) Such a holomorphic 2-form is usually denoted as $\omega_X$. It is unique up to a non-zero complex scalar. The irregularity $q$ of a surface $X$ is by definition equal to $h^{0,1}(X) = \dim H^{0,1}(X)$. In fact, switching the condition $q = 0$ to $H^1(X;\mathbb{Z}) = 0$ or even $\pi_1(X) = 0$ gives equivalent definitions (see proposition 2.7 and corollary 5.18).

Now we introduce two classes of examples of K3 surfaces.

**Example 2.3.** A non-singular quartic surface in $\mathbb{P}^3$ is a K3 surface. For example, the zero locus of $X^4 + Y^4 + Z^4 + W^4 = 0$ defines a K3 surface in $\mathbb{P}^3$.

*Proof.* Let $X$ be a non-singular quartic surface. By the adjunction formula,

$$K_X \cong (K_{\mathbb{P}^3} + X)|_X.$$

Let $H$ be the hyperplane section. We know that $K_{\mathbb{P}^3} \sim -4H$ and $X \sim 4H$ in $\mathbb{P}^3$. Therefore, $K_X$ is trivial.

Next, we will use the Lefschetz hyperplane theorem [Kon20, p.38] to show that $X$ has trivial first homology group and therefore has irregularity 0. Consider the degree 4 Veronese embedding

$$\nu_4 : \mathbb{P}^3 \longrightarrow \mathbb{P}^{34},$$
$$[X : Y : Z : W] \mapsto [X^4 : X^3Y : X^3Z : \cdots : W^4].$$

The coordinates of $\mathbb{P}^{34}$ are the degree 4 monomials in four variables. Therefore $X \subseteq \mathbb{P}^{34}$ is the intersection of $\mathbb{P}^3$ with a hyperplane $H_X$. By the Lefschetz hyperplane theorem,

$$H_1(X) = H_1(\mathbb{P}^3 \cap H_X) \cong H_1(\mathbb{P}^3),$$

which is trivial. Therefore, a non-singular quartic surface in $\mathbb{P}^3$ defines a $K3$ surface. □

Next we introduce the other more geometric examples, called Kummer surfaces.

**Definition 2.4.** Let $v_1, v_2, v_3, v_4$ be four vectors in $\mathbb{C}^2$ linearly independent over $\mathbb{R}$. Let $\Gamma$ be the free abelian group generated by $v_1, v_2, v_3, v_4$. It acts on $\mathbb{C}^2$ by translation. The quotient $A = \mathbb{C}^2/\Gamma$ is a complex manifold, called the 2-dimensional *complex torus*.

Multiplication by -1 defines an automorphism of $\mathbb{C}^2$ that is $\Gamma$-equivariant. Therefore it descends to an automorphism of the complex torus $A$, denoted as $-1_A$. The action of $-1_A$ is not free. Its fixed points correspond to $\frac{1}{2}\Gamma/\Gamma$, which is a set of 16 discrete points.

**Definition 2.5.** Let $A$ be a complex torus. The minimal resolution of $A/\{\pm 1_A\}$ is called a *Kummer surface*, denoted by $\mathrm{Km}(A)$.

More concretely, the Kummer surface associated with a complex torus $A$ can be built as follows. First blow up $A$ at the 16 points of $\frac{1}{2}\Gamma/\Gamma$, denoted as $\widetilde{A}$. Locally this is simply $\mathbb{C}^2$ blown up at the origin and is isomorphic to

$$\{((x, y), [u : v]) \in \mathbb{C}^2 \times \mathbb{P}^1 \mid xv = yu\}.$$

The automorphism $-1_{\widetilde{A}}$ acts on the $\mathbb{C}^2$ part:

$$((x, y), [u : v]) \mapsto ((-x, -y), [u : v]).$$

We can cover $\mathrm{Bl}_0(\mathbb{C}^2)$ by two affine charts $(U,(x_1,y_1))$ and $(V,(x_2,y_2))$. In local coordinates, the formula is

$$\sigma(x_1,y_1)=(x_1,x_1y_1),$$
$$\sigma(x_2,y_2)=(x_2y_2,y_2).$$

The two charts are identified by

$$x_1=x_2y_2,\qquad x_1y_1=x_2.$$

Now the automorphism $-1_{\widetilde{A}}$ induces

$$\nu:(x_1,y_1)\longmapsto(-x_1,y_1),\qquad (x_2,y_2)\longmapsto(x_2,-y_2).$$

Now $\widetilde{A}/\{\pm 1_A\}$ has local coordinates $(x_1^2,y_1)$ and $(x_2,y_2^2)$. Therefore it is smooth. The quotient $\widetilde{A}/-1_{\widetilde{A}}$ is the Kummer surface.

**Proposition 2.6.** *Kummer surfaces are K3 surfaces.*

*Proof.* Let $X=\mathrm{Km}(A)$ where $A$ is a complex torus. First we show that it admits a nowhere vanishing holomorphic 2-form. The form $dz_1\wedge dz_2$ is a non-vanishing holomorphic 2-form on $\mathbb{C}^2/\Gamma$ and is invariant under $-1_A$. We only need to care about what happens at the 16 fixed points.

In local coordinates,

$$dz_1\wedge dz_2=x_1\,dx_1\wedge dy_1=-x_2\,dx_2\wedge dy_2,$$

which is

$$d(x_1^2)\wedge dy_1=-d(x_2^2)\wedge dy_2,$$

and is non-vanishing. Because $-1_A$ fixes the exceptional curves, we only need that they are non-vanishing. Therefore we have constructed a non-vanishing holomorphic 2-form on $\mathrm{Km}(A)$.

To show that irregularity $q(X)=0$, we use the Noether's formula [Har77, p.363]:

$$\chi(\mathcal{O}_X)=\frac{1}{12}(c_1^2(X)+e(X)).$$

The left-hand side is

$$\chi(\mathcal{O}_X)=h^{0,0}(X)-h^{0,1}(X)+h^{0,2}(X)=2-q(X).$$

The right-hand side is

$$\frac{1}{12}(c_1^2(X)+e(X))=\frac{1}{12}((-K_X)^2+e(X))=\frac{1}{12}\,e(X).$$

Therefore it suffices to show that $e(X)=24$. Start with $e(A)=1-4+6-4+1=0$. A blow-up at a point replaces a point with $\mathbb{P}^1$, which adds 1 to the Euler characteristic. Therefore, $e(\widetilde{A})=16$. The map $\widetilde{A}\to X$ is a double-cover branched over 16 copies of $\mathbb{P}^1$. Outside of the branched locus, it is a normal 2-fold covering. Hence their Euler characteristics satisfy the following relation

$$e(\widetilde{A})-16e(\mathbb{P}^1)=2(e(X)-16e(\mathbb{P}^1)),$$
$$e(X)=\frac{1}{2}(e(\widetilde{A})+16e(\mathbb{P}^1)),$$
$$e(X)=24.$$

We have shown that $X=\mathrm{Km}(A)$ is a K3 surface. □

Now we turn to some basic properties of K3 surfaces.

**Proposition 2.7.** *Let $X$ be a K3 surface. Then $H_1(X;\mathbb{Z})=0$.*

*Proof.* Consider the exact exponential sheaf sequence

$$0 \longrightarrow \underline{\mathbb{Z}} \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_X^* \xrightarrow{exp} 0.$$

This induces a long exact sequence of cohomology groups

$$0 \longrightarrow H^0(X, \underline{\mathbb{Z}}) \longrightarrow H^0(X, \mathcal{O}_X) \longrightarrow H^0(X, \mathcal{O}_X^*) \longrightarrow H^1(X, \underline{\mathbb{Z}}) \longrightarrow H^1(X, \mathcal{O}_X) \longrightarrow \cdots$$

where $H^0(X, \underline{\mathbb{Z}}) \cong \mathbb{Z}$, $H^0(X, \mathcal{O}_X) \cong \mathbb{C}$ and $H^0(X, \mathcal{O}_X^*) \cong \mathbb{C}^\vee$ as global functions on a connected compact complex manifold must be constant. Moreover, $H^1(X, \mathcal{O}_X) \cong H^{0,1}(X)$ and is trivial. Therefore, $H^1(X, \underline{\mathbb{Z}}) \cong H^1(X; \mathbb{Z})$ is trivial, and so is $H_1(X; \mathbb{Z})$. □

**Proposition 2.8.** *Let $X$ be a K3 surface. Then $H^2(X; \mathbb{Z}) \cong \mathbb{Z}^{22}$.*

*Proof.* Using the Noether's formula as in the proof of proposition 2.6, we know $e(X) = 24$. Because $H_1(X; \mathbb{Z}) = 0$, the second betti number of a K3 surface is 22. Finally, by the universal coefficient theorem, $H^2(X; \mathbb{Z}) \cong \mathrm{Hom}(H_2(X; \mathbb{Z}), \mathbb{Z})$ and is free. Thus $H^2(X; \mathbb{Z}) \cong \mathbb{Z}^{22}$. □

From the above two propositions, we conclude that a K3 surface has the following Hodge diamond:

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 1 & & 20 & & 1 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$$

**Proposition 2.9.** *Every K3 surface is minimal.*

*Proof.* The canonical bundle is trivial so it is nef. Alternatively, one can use the adjunction formula or the Riemann-Roch theorem for surface to show that there is no curve on a K3 surface with self-intersection number -1. □

Next we state three facts about K3 surfaces.

**Theorem 2.10** ([Siu83])**.** *Every K3 surface is Kähler.*

**Theorem 2.11.** *All K3 surfaces are diffeomorphic.*

**Theorem 2.12.** *Not all K3 surfaces are projective.*

In this paper, we will assume the first result that every K3 surface is Kähler. We will prove the latter two results in later sections.

## 3. The second cohomology groups of K3 surfaces

The cup product gives $H^2(X; \mathbb{Z})$ a lattice structure. It turns out that this lattice structure captures much important information and is essential in the statement of the Torelli theorem for K3 surfaces. Therefore we first describe some results from general lattice theory.

**Definition 3.1.** Let $L$ be a free abelian group of rank $r$. Let

$$\langle \, , \, \rangle : L \times L \to \mathbb{Z}$$

be an integral-valued symmetric bilinear form. It is called non-degenerate if $\langle x, y \rangle = 0$ for all $y \in L$ implies $x = 0$. A *lattice* of rank $r$ is a pair $(L, \langle \, , \, \rangle)$ of a free abelian group of rank $r$ and a non-degenerate symmetric bilinear form.

**Definition 3.2.** A *lattice isomorphism* is a group isomorphism that preserves the bilinear form. That is, $f : L_1 \to L_2$ is a lattice isomorphism if it is linear and for any $x, y \in L_1$, $\langle x, y\rangle = \langle f(x), f(y)\rangle$. For a fixed lattice $L$, we denote the group of lattice automorphisms by $O(L)$ and sometimes call it the *orthogonal group* of $L$.

Fix a basis $\{e_i\}_{i=1}^r$ of $L$. For each $x \in L$, write $x = \Sigma_i x_i e_i$. Then $f(x) = \langle x, x\rangle$ is a quadratic form. By Sylvester's law of inertia, we can find a basis $\{t_i\}_{i=1}^r$ defined over $\mathbb{R}$ such that if $x = \Sigma_i t_i$, $f(x)$ can be expressed as

$$f(x) = t_1^2 + \cdots + t_p^2 - t_{p+1}^2 - \cdots - t_{p+q}^2, \quad p + q = r.$$

Moreover, $p, q$ do not depend on the choice of basis.

**Definition 3.3.** For a lattice $L$, the pair $(p, q)$ is called the *signature* of $L$. The difference $p - q$ is sometimes called the signature as well.

**Definition 3.4.** A lattice $L$ is called *positive-definite* or *negative-definite* if $p = r$ or $q = r$, respectively. It is called *indefinite* if $p, q > 0$.

**Definition 3.5.** A lattice $L$ is called *unimodular* if the associated matrix $A = (a_{ij})$ where $a_{ij} = \langle e_i, e_j\rangle$ for a basis $\{e_i\}_{i=1}^r$ has determinant $\pm 1$. It is clear that this does not depend on the basis chosen.

**Definition 3.6.** A lattice $L$ is called *even* if the norm of any element $x^2 = \langle x, x\rangle$ is even. It is called *odd* if it is not even.

**Definition 3.7.** The lattice $L(m)$ is defined to be $(L, m\langle\, ,\rangle)$ where $m \in \mathbb{Z}$.

The following classification theorem is very useful in the study of K3 surfaces. We will use it to show that the second cohomology group of K3 surfaces are all isomorphic to a fixed abstract lattice.

**Theorem 3.8.** *Let $L$ be an indefinite even unimodular lattice. Then the signature $p - q$ is divisible by 8. Moreover,*

*(1) if $p \le q$, then $L \cong U^{\oplus p} \oplus E_8^{\oplus(q-p)/8}$;*
*(2) if $p \ge q$, then $L \cong U^{\oplus q} \oplus E_8(-1)^{\oplus(p-q)/8}$.*

The lattice $U$ is called the hyperbolic lattice, and has signature $(1, 1)$ and associated matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

The lattice $E_8$ is a negative-definite lattice of rank 8. Its associated matrix can be expressed as

$$\begin{pmatrix} -2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -2 \end{pmatrix}$$

The name of the lattice comes from the associated Dynkin diagram. See [Kon20, Ch.1] for more details and the proof of the above theorem.

**Theorem 3.9.** *Let $X$ be a K3 surface. Then $H^2(X; \mathbb{Z})$ with the cup product pairing is an even unimodular lattice with signature $(3, 19)$ and is therefore isomorphic to the abstract lattice*

$$U^{\oplus 3} \oplus E_8^{\oplus 2}.$$

*Proof.* By Poincare duality, the cup product pairing is non-degenerate and unimodular. As a consequence of Wu's formula, it is also an even lattice (see [Kon20, p.57]). By the Hirzebruch signature theorem [MS74, Ch.19], the signature of $X$ is equal to $1/3$ of the first Pontryagin class. That is,

$$\sigma(X) = \frac{1}{3}p_1(X) = \frac{1}{3}\big(c_1^2(X) - 2c_2(X)\big) = -16.$$

Therefore, the signature of $H^2(X;\mathbb{Z})$ is $(3,19)$. By the classification theorem 3.8, it is isomorphic to $U^{\oplus 3} \oplus E_8^{\oplus 2}$. □

**Remark 3.10.** Applying the Hirzebruch signature theorem gives us the correct answer very quickly but it may seem a bit mysterious why the signature is $(3,19)$. Another way to see this is the following. $H^{2,0}(X;\mathbb{R})$ and $H^{0,2}(X;\mathbb{R})$ are spanned by $\mathrm{Re}(\omega_X)$ and $\mathrm{Im}(\omega_X)$. The non-vanishing holomorphic 2-form $\omega_X$ satisfies the following *Riemann condition*

$$\langle \omega_X, \omega_X \rangle = 0, \langle \omega_X, \overline{\omega_X} \rangle > 0.$$

Then by a direct calculation, we have

$$\langle \mathrm{Re}(\omega_X), \mathrm{Re}(\omega_X) \rangle = \langle \mathrm{Im}(\omega_X), \mathrm{Im}(\omega_X) \rangle > 0, \langle \mathrm{Re}(\omega_X), \mathrm{Im}(\omega_X) \rangle = 0.$$

Therefore $H^{2,0}(X;\mathbb{R}) \oplus H^{0,2}(X;\mathbb{R})$ is a positive-definitive space and contribute signature $(2,0)$. Finally, by the Hodge index theorem, $H^{1,1}(X;\mathbb{R})$ have signature $(1,19)$: pick one positive-definitive class, e.g. the Kähler class, then the orthogonal complement is negative-definitive. Hence $H^2(X;\mathbb{R}) = H^{2,0}(X;\mathbb{R}) \oplus H^{1,1}(X;\mathbb{R}) \oplus H^{0,2}(X;\mathbb{R})$ has signature $(3,19)$.

**Definition 3.11.** Let $L, S$ be lattices. A linear map from $S$ to $L$ preserving the bilinear form is called an *embedding* of lattices. An embedding $S \subset L$ of lattices is called *primitive* if the quotient $L/S$ is torsion-free.

The following result about embedding of lattice will be used in the proof of the Torelli theorem for K3 surfaces. We state it here for future reference.

**Proposition 3.12.** *Let $L$ be an even unimodular lattice and assume that $L$ has an orthogonal decomposition $L = U^{\oplus k} \oplus K$. Then every even lattice of rank less than or equal to $k$ can be primitively embedded into $L$, meaning that the quotient lattice is torsion-free. This result also holds when the lattices are degenerate.*

*Proof.* See [Kon20, p.24]. □

Next we turn our attention to the following sublattice.

**Definition 3.13.** For a K3 surface $X$, define

$$S_X = H^{1,1}(X;\mathbb{R}) \cap H^2(X;\mathbb{Z})$$

to be the *Néron-Severi lattice* of $X$. Its orthogonal complement in $H^2(X;\mathbb{Z})$ is called the *transcendental lattice* and is denoted as $T_X$.

By the Lefschetz theorem on $(1,1)$-classes, every element in $S_X$ is represented by a divisor. Conversely the Chern class of any divisor must lie in $H^{1,1}(X;\mathbb{R}) \cap H^2(X;\mathbb{Z})$. So $S_X$ encodes information about all divisors on $X$.

**Remark 3.14.** While $H^{1,1}(X;\mathbb{R})$ is a vector space over $\mathbb{R}$ of rank 20 and $H^2(X;\mathbb{Z})$ is a free abelian group of rank 22, it is important to notice that $S_X$ need *not* be a free abelian group of rank 20. In fact, it can have any rank between 0 and 20. One may think of $H^2(X;\mathbb{Z})$ as the integral points of $H^2(X;\mathbb{R})$ and $H^{1,1}(X;\mathbb{R})$ as a co-dimension 2 vector subspace. Just as the line $y = \sqrt{3}x$ in $\mathbb{R}^2$ does not contain any integral points other than the origin, $H^{1,1}(X;\mathbb{R})$ may contain no integral points other than the origin or only 'some' integral points. If $S_X$ only contains the origin, then $X$ is not

projective as if it is, the hyperplane section is a non-zero element in $S_X$. A more precise result is given in proposition 3.21.

**Proposition 3.15.** *Let $\delta \in S_X$ with $\delta^2 \geq -2$. Then $\delta$ or $-\delta$ is represented by an effective divisor.*

*Proof.* We use the Riemann-Roch theorem for surfaces. Let $L$ be the associated divisor with $\delta$. Then

$$\chi(\mathcal{O}(L)) = \frac{1}{2}L \cdot (L - K) + \chi(\mathcal{O}_X).$$

Thus

$$\dim H^0(X, \mathcal{O}(L)) - \dim H^1(X, \mathcal{O}(L)) + \dim H^2(X, \mathcal{O}(L)) = \frac{1}{2}\delta^2 + 2,$$

so

$$\dim H^0(X, \mathcal{O}(L)) + \dim H^2(X, \mathcal{O}(L)) \geq 1 + \dim H^1(X, \mathcal{O}(L)) \geq 1.$$

By Serre duality, $\dim H^2(X, \mathcal{O}(L)) = \dim H^0(X, \mathcal{O}(-L))$. Therefore, at least one of $\dim H^0(X, \mathcal{O}(L))$ and $\dim H^0(X, \mathcal{O}(-L))$ is positive. Either $\delta$ or $-\delta$ is represented by an effective divisor. □

On the other hand, let $C$ be a non-singular irreducible curve on $X$. By the adjunction formula $2g(C) - 2 = K_X \cdot C + C^2 = C^2$, we have $C^2 \geq -2$. Moreover, if $C^2 = -2$, $C$ is a rational curve.

Next we study the question of where do the Kähler classes sit inside $H^2(X; \mathbb{R})$. Maybe surprisingly, there is a very nice description of this. A Kähler class $\kappa$ satisfies the following properties:

(1) $\kappa \in H^{1,1}(X; \mathbb{R})$;
(2) $\langle \kappa, \kappa \rangle > 0$;
(3) $\langle \kappa, c \rangle > 0$ for any $c \in H^{1,1}(X; \mathbb{R})$ that is the Chern class of an effective divisor.
(4) Positive linear combination of Kähler classes is also a Kähler class.

The third condition is equivalent to $\langle \kappa, c \rangle > 0$ for any $c$ that is the Chern class of an irreducible curve. We first consider the first two conditions.

**Definition 3.16.** Let $X$ be a K3 surface. Define

$$P(X) = \{x \in H^{1,1}(X; \mathbb{R}) \mid \langle x, x \rangle > 0\}.$$

It has two connected components, and we call the one that contains a Kähler class the *positive cone* of $X$, denoted as $P^+(X)$.

Note that $H^{1,1}(X; \mathbb{R})$ has signature $(1, 19)$. The norm may be written as $f(x) = x_1^2 - x_2^2 - \cdots - x_{20}^2$. This is why $P(X)$ has two connected components. The Kähler classes must lie in one of the connected component, but not both. If there are two Kähler classes $x, y$, one in each of the connected component, we may scale it such that $x_1 = -y_1$. Taking the sum will give a Kähler class with first component 0, which is impossible.

Next we consider condition (3). We first have the following lemma.

**Lemma 3.17.** *Let $x \in P^+(X), y \in \overline{P^+(X)}, y \neq 0$. Then $\langle x, y \rangle > 0$.*

*Proof.* By assumptions,

$$x_1^2 > \sum_{i=2}^{20} x_i^2, \ y_1^2 \geq \sum_{i=2}^{20} y_i^2.$$

By the Cauchy-Schwarz inequality, we have

$$x_1^2 y_1^2 > (\sum_{i=2}^{20} x_i^2)(\sum_{i=2}^{20} y_i^2) \geq (\sum_{i=2}^{20} x_i y_i)^2.$$

This proves the claim. □

Therefore, $\langle \kappa, c\rangle > 0$ for any curve $c$ with $c^2 \geq 0$. We only need to care about curves with self-intersection number -2. Let $\delta$ be such a curve, and consider $H_\delta = \{x \in H^{1,1}(X;\mathbb{R}) \mid \langle x, \delta\rangle = 0\}$. The space $H^{1,1}(X;\mathbb{R})\backslash H_\delta$ has two connected components: one of them contains vectors with positive inner product with $\delta$; the other contains those with negative inner product. The Kähler class must lie in the connected component where every element has positive inner product with every $\delta$. This leads to the following definition.

**Definition 3.18.** A *chamber* is a connected component of the complement of the positive cone of the union of hyperplanes

$$P^+(X)\backslash \bigcup_{\delta^2=-2} H_\delta.$$

The chamber that contains a Kähler class is called the *Kähler cone*, denoted as $\mathcal{D}(X)$.

Again, only one of the chamber may contain a Kähler class. If there exists a Kähler class in each of two chambers, denoted as $\kappa_1$ and $\kappa_2$, then $\kappa_t = t\kappa_1 + (1-t)\kappa_2$ is also a Kähler class for any $t$. But for some $t$, $\kappa_t$ will lie on one of the hyperplanes, which is impossible.

The chambers are in fact fundamental domains of certain reflection group.

**Definition 3.19.** For $\delta \in S_X$ with $\delta^2 = -2$, consider the following map on $H^{1,1}(X;\mathbb{R})$:

$$s_\delta(x) = x + \langle x, \delta\rangle \delta.$$

The map $s_\delta$ preserves the bilinear form and satisfies $s_\delta^2 = 1$. Let $W$ be the subgroup of $O(H^{1,1}(X;\mathbb{R}))$ generated by such $s_\delta$. This is called the *reflection group*.

**Theorem 3.20.** *Any chamber $C$ is a fundamental domain of the actions of $W$ on $P^+(X)$.*

*Proof.* See [Kon20, p.30]. □

Finally we give a criterion for when a K3 surface is projective which will be used later.

**Proposition 3.21.** *A K3 surface $X$ is projective if and only if $S_X \cap \mathcal{D}(X)$ is non-empty.*

*Proof.* By Nakai's criterion [Kon20, p.38], a divisor $D$ is ample if and only if $D^2 > 0$ and $D \cdot C > 0$ for any irreducible curve $C$. Thus any element in $S_X \cap \mathcal{D}(X)$ is an ample divisor. Conversely, the hyperplane section of a projective K3 surface lies in $S_X \cap \mathcal{D}(X)$. □

## 4. Complex tori and Kummer surfaces

In this section we study various properties of complex tori and Kummer surfaces and prove the Torelli theorems for them.

Unlike K3 surfaces, deciding when two complex tori are isomorphic (as complex surfaces) is relatively easy as they are quotients of $\mathbb{C}^2$. Suppose there is an isomorphism between $\mathbb{C}^2/\Gamma$ and $\mathbb{C}^2/\Gamma'$. It lifts to a biholomorphism of $\mathbb{C}^2$, which is a linear map, that takes $\Gamma$ to $\Gamma'$. Conversely, any such biholomorphism descends to an isomorphism between the two complex tori. This formulation, however, is not quite the same as the one for K3 surfaces: it involves the 'ambient space' $\mathbb{C}^2$. We are more interested in just the cohomology groups themselves. We can ask the following question: let $\psi : \Gamma \to \Gamma'$ be an isomorphism of groups. When is this isomorphism induced by an isomorphism of the complex tori?

This is very similar to our previous discussion on elliptic curves, which are in a sense '1-dimensional' version of complex tori. Let $A = V/\Gamma$ be a complex torus. There is a natural isomorphism

$$V^* \cong H^0(A, \Omega_A^1), \quad V \cong H^0(A, \Omega_A^1)^*.$$

This is because any global holomorphic 1-form on $A$ lifts to one on $V$ which is periodic along the four directions of $\Gamma$. Any such form must be of the form $\omega = c_1 dz_1 + c_2 dz_2$. The form gives a linear map

$$v \mapsto \int_v \omega.$$

That the 1-form $\omega$ being translation-invariant guarantees that this is a linear map. This establishes the above isomorphism. Next consider

$$\psi^* \otimes \mathbb{C} : H^1(A'; \mathbb{C}) \to H^1(A; \mathbb{C}).$$

By the Hodge decomposition, $H^1(A; \mathbb{C}) \cong H^0(A, \Omega^1_A) \oplus H^1(A, \mathcal{O}_A)$. Therefore, we may conclude that $\psi$ is induced by an isomorphism of complex manifolds if and only if the associated map on $H^1(A; \mathbb{C})$ preserves the Hodge decomposition: it induces an isomorphism on $H^0(A, \Omega^1_A)$. We formulate this into the following theorem.

**Theorem 4.1.** *Let $A = V/\Gamma$, $A' = V'/\Gamma'$ be two complex tori. Let*

$$\psi : \Gamma^* \longrightarrow \Gamma'^*$$

*be an isomorphism of groups. If $\psi(H^0(A, \Omega^1_A)) = H^0(A', \Omega^1_{A'})$, then there exists an isomorphism $\varphi : A' \to A$ of complex manifolds such that $\varphi^* = \psi$.*

This already resembles the Torelli theorem we are trying to prove for K3 surfaces. The next step is to prove a Torelli theorem for Kummer surfaces with the help of this result. However, Kummer surfaces, as they are K3 surfaces, have trivial first homology groups. No information can be extracted from the first cohomology group. Therefore, it is necessary to formulate a Torelli theorem for complex tori considering when an isomorphism between the second cohomology groups is induced by an isomorphism of complex tori. This is the following result:

**Theorem 4.2** (Torelli Theorem for Complex Tori)**.** *Let $A = V/\Gamma$, $A' = V'/\Gamma'$ be two complex tori. Let*

$$\phi : H^2(A; \mathbb{Z}) \longrightarrow H^2(A'; \mathbb{Z})$$

*be an isomorphism of lattices such that*

*(1) $(\phi \otimes \mathbb{C})(H^0(A, \Omega^2_A)) = H^0(A', \Omega^2_{A'})$;*
*(2) there exists an isomorphism $\psi : \Gamma^* \to \Gamma'^*$ such that $\phi = \pm\psi \wedge \psi$.*

*Then there exists a unique isomorphism $\varphi : A' \to A$ of complex manifolds such that $\varphi^* = \phi$.*

The proof of this result uses the fact that $H^2(A; \mathbb{Z}) \cong \bigwedge^2 H^1(A; \mathbb{Z})$ and shows that $\phi$ descends to an isomorphism on $H^1(A; \mathbb{Z})$ that satisfies the conditions in the previous result.

*Proof.* We have

$$V^* \cong H^0(A, \Omega^1_A), \qquad V \cong \left(H^0(A, \Omega^1_A)\right)^*, \qquad \Gamma \cong H_1(A; \mathbb{Z}).$$

We are going to show that $\phi$ induces isomorphisms on

$$H^0(A', \Omega^1_{A'})^* \longrightarrow H^0(A, \Omega^1_A)^*$$

and

$$H_1(A'; \mathbb{Z}) \longrightarrow H_1(A; \mathbb{Z}).$$

By condition (2), $\phi = \pm\psi \wedge \psi$ where $\psi : H^1(A; \mathbb{Z}) \to H^1(A'; \mathbb{Z})$ is an isomorphism. We want to show that $\psi$ fixes $H^0(A, \Omega^1_A)$ as a summand of $H^1(A; \mathbb{C})$ by using condition (1), and therefore show that $\psi$ induces an isomorphism on $H^0(A, \Omega^1_A)^*$. We abuse the notation by letting $\psi$ denote $\psi \otimes \mathbb{C} : H^1(A; \mathbb{C}) \to H^1(A'; \mathbb{C})$. Let $\omega_1, \omega_2$ be two basis of $H^0(A, \Omega^1_A)$. Let

$$\psi(\omega_1) = \eta_1 + \bar{\tau}_1, \qquad \psi(\omega_2) = \eta_2 + \bar{\tau}_2,$$

where $\eta_1, \tau_1, \eta_2, \tau_2 \in H^0(A', \Omega^1_{A'})$. Then

$$(\psi \wedge \psi)(\omega_1 \wedge \omega_2) = (\eta_1 + \bar{\tau}_1) \wedge (\eta_2 + \bar{\tau}_2).$$

By condition (1),

$$(\eta_1 + \bar{\tau}_1) \wedge (\eta_2 + \bar{\tau}_2) \in H^0(A', \Omega^2_{A'}).$$

Thus

$$\eta_1 \wedge \bar{\tau}_2 + \bar{\tau}_1 \wedge \eta_2 = 0, \qquad \bar{\tau}_1 \wedge \bar{\tau}_2 = 0.$$

If $\tau_1 \neq 0$, then $\tau_2 = a\tau_1$. Then

$$\bar{a}\, \eta_1 \wedge \bar{\tau}_1 + \bar{\tau}_1 \wedge \eta_2 = 0,$$

and hence

$$(\bar{a}\eta_1 + \eta_2) \wedge \bar{\tau}_1 = 0.$$

Therefore,

$$\bar{a}\eta_1 + \eta_2 = b\bar{\tau}_1.$$

This can only happen if $b = 0$, so $\eta_2 = -\bar{a}\eta_1$. In this case,

$$(\psi \wedge \psi)(\omega_1 \wedge \omega_2) = 0.$$

This contradicts that $\phi$ is an isomorphism. Therefore $\tau_1 = 0$. By symmetry, $\tau_2 = 0$. The map $\psi$ restricts to an isomorphism

$$H^0(A, \Omega^1_A) \longrightarrow H^0(A', \Omega^1_{A'}).$$

This gives an isomorphism

$$\varphi : H^0(A', \Omega^1_{A'})^* / H_1(A'; \mathbb{Z}) \longrightarrow H^0(A, \Omega^1_A)^* / H_1(A; \mathbb{Z}).$$

And by construction, $\varphi^* = \phi$. It is clear that $\varphi$ is unique as any automorphism of a complex torus $A$ that induces the identity map on $H^2(A; \mathbb{Z})$ also induces the identity map on $H_1(A; \mathbb{Z})$ and $H^0(A, \Omega^1_A)$. □

We record the following technical lemma for later use in this section.

**Proposition 4.3.** *The second condition in theorem 4.2 is equivalent to the condition that there exists an isomorphism $\psi_2 : \Gamma^* \otimes \mathbb{F}_2 \to \Gamma'^* \otimes \mathbb{F}_2$ such that $\phi \bmod 2 = \psi_2 \wedge \psi_2$.*

*Proof.* See [Kon20, p.71]. □

Next we turn to the Torelli theorem for Kummer surfaces. We first introduce a characterization of Kummer surfaces. As a Kummer surface is obtained by blowing up 16 points on a complex torus and taking a quotient, it contains 16 exceptional rational curves. We will prove a converse. First we need the following result.

**Proposition 4.4.** *Let $M$ be a compact complex manifold and let $D$ be a non-singular effective divisor. Then the following are equivalent:*

*(1) There exists a double cover $\pi : \widetilde{M} \to M$ branched over $D$.*
*(2) $D' = \frac{1}{2}D \in \mathrm{Pic}(M)$.*

*Moreover,*

$$K_{\widetilde{M}} \cong \pi^*(K_M \otimes \mathcal{O}(D')).$$

*Proof.* (1) ⇒ (2): Locally on $M$, around every point of $D$, choose a local coordinate $(z_1, \ldots, z_n)$ such that locally $D = \{z_1 = 0\}$. We may choose local coordinates $(w_1, \ldots, w_n)$ on $\widetilde{M}$ such that $z_1 = w_1^2$ and $z_i = w_i$ $(i \geq 2)$.

On another coordinate chart $(z_1', z_2', \ldots, z_n')$, suppose $D = \{z_1' = 0\}$. Then $z_1' = gz_1$ where $g$ is a non-vanishing holomorphic function. Then we get $(w_1')^2 = gw_1^2$ on $\widetilde{M}$. Now $g$ must have a

non-vanishing holomorphic square root. The collection of such $g$ defines a line bundle $L$ on $M$ such that $L^{\otimes 2} = D$.

(2) $\Rightarrow$ (1): Let $p : L \to M$ be a line bundle such that $L^{\otimes 2} = D$. Let $s$ be a section of $p^{\otimes 2}$ such that $(s) = D$. Then

$$\widetilde{M} = \{y \in L \mid y^{\otimes 2} = s(p(y))\},$$

is the desired double cover.

To show the last statement, we still compute in local coordinates. A holomorphic $n$-form on $M$ is $dz_1 \wedge dz_2 \wedge \cdots \wedge dz_n$. Pulling it back to $\widetilde{M}$ gives

$$\pi^*(dz_1 \wedge dz_2 \wedge \cdots \wedge dz_n) = dw_1 \wedge dw_2 \wedge \cdots \wedge dw_n = 2z_1 dz_1 \wedge dz_2 \wedge \cdots \wedge dz_n$$

where it vanishes along $\{z_1 = 0\} = D$. Therefore

$$K_{\widetilde{M}} \cong \pi^* K_M \otimes \mathcal{O}(R)$$

where $2R = \pi^* D$.

Then because $R = \pi^* D'$, we get

$$K_{\widetilde{M}} \cong \pi^*(K_M \otimes \mathcal{O}(D'))$$

as desired. □

**Proposition 4.5.** *Let $X$ be a K3 surface containing 16 disjoint non-singular rational curves $E_1, \ldots, E_{16}$. Assume that*

$$\frac{1}{2}\sum_{i=1}^{16} E_i \in S_X.$$

*Then there exists a unique complex torus $A$ up to isomorphism such that $X$ is the Kummer surface associated with $A$ and $\{E_i\}_{i=1}^{16}$ are the exceptional curves.*

*Proof.* By proposition 4.4, there is a double cover of $X$ branched over the sixteen curves $\pi : \widetilde{X} \to X$. Let $\widetilde{E}_i = \pi^{-1}(E_i)$. Then

$$-4 = 2(E_i)^2 = (\pi^* E_i)^2 = 4\widetilde{E}_i^2,$$
$$\widetilde{E}_i^2 = -1.$$

Therefore, the pre-image of these sixteen curves each has self-intersection number $-1$. Therefore we can perform the blow down action on the sixteen curves $\sigma : \widetilde{X} \to Y$. We claim that $Y$ is the complex torus.

First we show that $Y$ has trivial canonical bundle. We know that $K_X = 0$ by our assumption. By the previous proposition,

$$K_{\widetilde{X}} \cong \pi^* K_X \otimes \mathcal{O}\left(\sum_{i=1}^{16} \widetilde{E}_i\right),$$
$$K_{\widetilde{X}} \cong \mathcal{O}\left(\sum_{i=1}^{16} \widetilde{E}_i\right).$$

Next, for the blow down procedure, locally for a blow-up at a point $p$, we have

$$K_{\mathrm{Bl}_p X} \cong K_X + E_p$$

where $E_p$ is the exceptional divisor. Therefore,

$$K_Y \cong K_{\widetilde{X}} - \sum_{i=1}^{16} \widetilde{E}_i,$$
$$K_Y \cong \mathcal{O}_Y.$$

Second, we compute the Euler characteristic. The branched cover gives

$$\chi(\widetilde{X}) - 16\chi(\mathbb{P}^1) = 2(\chi(X) - 16\chi(\mathbb{P}^1)),$$
$$\chi(\widetilde{X}) = 16.$$

Then each blow-down decreases the Euler characteristic by 1. Therefore $\chi(Y) = 0$. We have shown that $Y$ is a complex torus as claimed.

If $\mathrm{Km}(A) \cong \mathrm{Km}(A')$ with the same sixteen curves, then there is an isomorphism from $A/-1_A$ to $A'/-1_{A'}$, which means $A$ and $A'$ are isomorphic. □

Now we state the main theorem for this section.

**Theorem 4.6** (Torelli Theorem for Kummer Surfaces)**.** *Let $X$ and $X'$ be K3 surfaces and assume that $X$ is a projective Kummer surface. Suppose there is an isomorphism*

$$\phi : H^2(X;\mathbb{Z}) \longrightarrow H^2(X';\mathbb{Z})$$

*of lattices. We abuse the notation by letting $\phi$ also denote $\phi \otimes \mathbb{C}$. If*

*(1) $\phi(\omega_X) \in \mathbb{C}\omega_{X'}$;*
*(2) $\phi(\mathcal{D}(X)) = \mathcal{D}(X')$.*

*Then there exists a unique isomorphism $\varphi : X' \to X$ of complex manifolds with $\varphi^* = \phi$.*

For usage in the later argument of the Torelli theorem for general K3 surfaces, we only assume one of the K3 surfaces is a (projective) Kummer surface. The reason we also assume that $X$ is projective is because we need the following lemma:

**Lemma 4.7.** *Let $X, X'$ be projective K3 surfaces. Suppose there is an isomorphism of lattices $\phi : H^2(X;\mathbb{Z}) \to H^2(X';\mathbb{Z})$ that satisfies $\phi(\omega_X) \in \mathbb{C}\omega_{X'}$. Then the following conditions are equivalent:*

*(1) $\phi$ sends any effective divisor to an effective divisor.*
*(2) $\phi$ sends any ample divisor to an ample divisor.*
*(3) $\phi$ sends any Kähler class to a Kähler class.*
*(4) $\phi(\mathcal{D}(X)) = \mathcal{D}(X')$.*

*Proof.* The equivalence of (3) and (4) is because the Kähler cone is a fundamental domain with respect to a reflection group (see theorem 3.20). The equivalence of (2) and (4) is because $\mathcal{D}(X) \cap S_X$ is the set of ample classes. Because $\phi$ sends $\omega_X$ to a multiple of $\omega_{X'}$, it preserves $H^{1,1}(X)$ and $S_X$. If $\phi$ sends any effective divisor to an effective divisor, then it maps the Kähler cone to the Kähler cone as any Kähler class must have positive inner product with effective divisors. Thus (1) implies (4). Finally, we show (2) implies (1). It suffices to show that $\phi(C)$ is an effective divisor for any irreducible curve $C$. As $C^2 \geq -2$, $\phi(C)^2 \geq -2$ and either $\phi(C)$ or $-\phi(C)$ is an effective divisor. Take an ample divisor $D$ in $X$. Then $\langle \phi(C), \phi(D) \rangle = \langle C, D \rangle > 0$ where $\phi(D)$ is an ample divisor by assumption. Thus $\phi(C)$ must be the effective divisor. □

**Lemma 4.8.** *Under the assumption of theorem 4.6, $X'$ is also a projective Kummer surface.*

*Proof.* Because $\phi$ sends $S_X$ to $S_{X'}$ and sends $D(X)$ to $\mathcal{D}(X')$, $\mathcal{D}(X') \cap S_{X'}$ is non-empty. By proposition 3.21, $X'$ is projective. By lemma 4.7, $\phi$ takes effective divisors to effective divisors. For any $i$, $\phi(E_i)$ is an effective divisor. Suppose $\phi(E_i) = \Sigma_j m_j C_j$ where $m_j \in \mathbb{Z}^+$ and $C_j$ are irreducible curves. Then $E_i = \Sigma_j m_j \phi^{-1}(C_j)$. Because $\phi^{-1}$ also maps the Kähler cone to the Kähler cone, $\phi^{-1}(C_j)$ is effective. If $\phi(E_i)$ is reducible, this implies that $\dim H^0(X, \mathcal{O}_X(E_i)) \geq 2$, which contradicts $\dim H^0(X, \mathcal{O}_X(E_i)) = 1$. Thus $\phi(E_i)$ is an irreducible effective divisor with self-intersection number -2, and is therefore a rational curve. We conclude that there exist 16 mutually disjoint non-singular rational curves $\phi(E_1), \ldots \phi(E_{16})$ on $X'$ such that $\frac{1}{2}\Sigma_{i=1}^{16} \phi(E_i) \in S_{X'}$. By proposition 4.5, the K3 surface $X'$ is a projective Kummer surface. □

Let $A$ be a complex torus, and $X = \mathrm{Km}(A)$ be the associated Kummer surface. We first establish a relation between $H^2(A;\mathbb{Z})$ and $H^2(X;\mathbb{Z})$. In fact, $H^2(A;\mathbb{Z})$ embeds in $H^2(X;\mathbb{Z})$ as a primitive sublattice. We describe this below.

Let $\langle E_1, ..., E_{16}\rangle$ be the sublattice of $H^2(X;\mathbb{Z})$ generated by the classes of the 16 exceptional rational curves. Let $\Pi$ be the primitive sublattice of it in $H^2(X;\mathbb{Z})$, which is

$$\Pi := \{\Sigma_{i=1}^{16} a_i E_i \in H^2(X;\mathbb{Z}) \mid a_i \in \mathbb{Q}\} \subset S_X.$$

Let $\Pi^\perp$ denote the orthogonal complement of $\Pi$ in $H^2(X;\mathbb{Z})$. By a computation, it can be shown that $\det(\Pi^\perp) = -2^6$ [Kon20, p.98]. Let $\sigma : \widetilde{A} \to A$ be the blow-up of 16 points in $A$. Let $\pi : \widetilde{A} \to X$ be the branched double cover. Consider the homomorphism

$$q_* := \pi_* \circ \sigma^* : H^2(A;\mathbb{Z}) \to H^2(X;\mathbb{Z}).$$

**Proposition 4.9.** *The map $q_*$ is injective. Its image is equal to $\Pi^\perp$. Moreover, as a lattice, $\Pi^\perp$ is isomorphic to $U(2)^{\oplus 3}$.*

*Proof.* For $x \in H^2(A;\mathbb{Z})$, let $\iota$ be the non-trivial deck transformation of the covering $\pi$. Then

$$\pi^*\pi_*(\sigma^*(x)) = \sigma^*(x) + \iota^*(\sigma^*(x)) = 2\sigma^*(x).$$

Therefore, for $x, y \in H^2(A;\mathbb{Z})$,

$$4\langle \sigma^*(x), \sigma^*(y)\rangle = \langle \pi^*\pi_*(\sigma^*(x)), \pi^*\pi_*(\sigma^*(y))\rangle = 2\langle \pi_*(\sigma^*(x)), \pi_*(\sigma^*(y))\rangle.$$

Thus by choosing $x, y$ to avoid the 16 blown-up points, we have that for any $x, y \in H^2(A;\mathbb{Z})$,

$$\langle q_*(x), q_*(y)\rangle = 2\langle x, y\rangle.$$

This shows that $q_*$ is injective. Moreover, for any exceptional curve $E_i$,

$$2\langle \pi_*(\sigma^*(x)), E_i\rangle = \langle \pi^*\pi_*(\sigma^*(x)), \pi^*(E_i)\rangle = 2\langle \sigma^*(x), \pi^*(E_i)\rangle = 0.$$

This shows that the image of $q_*$ lies in $\Pi^\perp$. Because $H^2(A;\mathbb{Z})$ is isomorphic to $U^{\oplus 3}$ and $q_*$ multiplies the intersection form by 2, the image lattice has determinant $-2^6$. Therefore, $q_*(H^2(A;\mathbb{Z})) = \Pi^\perp$, which is isomorphic to the lattice $U(2)^{\oplus 3}$. □

Next we study the configuration of the 16 exceptional rational curves. Note that the 16 fixed points on the complex torus $A$ is in bijection with $\frac{1}{2}\Gamma/\Gamma$, and we have

$$\frac{1}{2}\Gamma/\Gamma = \frac{1}{2}H_1(A;\mathbb{Z})/H_1(A;\mathbb{Z}) \cong H_1(A;\mathbb{Z}/2\mathbb{Z}).$$

Therefore it has a vector space, and more generally, an affine space structure.

**Definition 4.10.** An $n$-dimensional *affine space* $V$ over a field $K$ is a set with an underlying vector space structure over $K$ (but forgetting the origin). The $k$-dimensional *subspaces* of the affine space are the $k$-dimensional vector subspaces and their translations by a vector $x \in V$. An $(n-1)$-dimensional subspace is called a *hyperplane*.

We denote the affine space $\frac{1}{2}\Gamma/\Gamma$ of $A$ by $I$. We have the following key lemma.

**Lemma 4.11.** *Let $X, X'$ be Kummer surfaces. Suppose there is a lattice isomorphism $\phi : H^2(X;\mathbb{Z}) \to H^2(X';\mathbb{Z})$ satisfying*

*(1) $\phi(\omega_X) \in \mathbb{C}\omega_{X'}$;*
*(2) $\phi$ preserves the 16 exceptional curves.*

*Then $\phi$ induces an affine map between the affine spaces $I, I'$.*

To prove this result, we need to study a certain geometry property of the exceptional curves. Let $E_1, ..., E_{16}$ be the 16 exceptional curves on the Kummer surface $X$. Let $J = \{1, ..., 16\}$. Define

$$Q_X = \{K \subset J \,|\, \frac{1}{2}\sum_{i\in K} E_i \in S_X\}.$$

By proposition 4.4, $J \in Q_X$. Moreover, by definition, $Q_X$ is closed under symmetric difference.

**Lemma 4.12.** *If $K \in Q_X$, then $|K|$ is equal to 0,8, or 16.*

*Proof.* For $K \neq \emptyset$, consider the double cover $\pi : \widetilde{X} \to X$ branched over $\Sigma_{i\in K} E_i$. Let $\widetilde{E_i}$ be the inverse image of $E_i$. They have self-intersection number $-1$ and we perform a blow-down $\sigma : \widetilde{X} \to Y$. By the argument in proposition 4.5 about the relation between the canonical bundle, $K_Y$ is trivial. Therefore by the classification of complex surfaces, $Y$ is either a complex torus or K3 surface, and $e(Y) = 0$ or 24. This shows that $|K| = 8$ or 16. □

**Lemma 4.13.** *Let $X$ be a Kummer surface. Then $Q_X$ consists of $\emptyset, J$, and hyperplanes.*

*Proof.* For a hyperplane $H$, if necessary switching to $J\backslash H$ such that the complement is a vector subspace, one can directly prove that there exists a K3 surface doubly-covering $X$ branched over $\Sigma_{i\in J\backslash H} E_i$. Conversely, for any $K \in Q_X$ with $|K| = 8$, it can be shown to be a hyperplane using a similar process. For details, see [Kon20, p.96]. □

*Proof of lemma 4.11.* By assumption, $\phi(S_X) = S_{X'}$. Hence $\phi(Q_X) = Q_{X'}$. This implies that $\phi$ takes hyperplanes to hyperplanes. If necessary by composing it with a translation, we may assume that $\phi$ preserves the origin and preserves 3-dimensional vector subspaces. Then $\phi$ also preserves their intersections, which are 2-dimensional vector subspaces. Thus if $\phi$ sends $\{0, x, y, x + y\}$ to $\{0, \phi(x), \phi(y), \phi(x + y)\}$, we have $\phi(x + y) = \phi(x) + \phi(y)$ and therefore $\phi$ is a linear map. □

**Proposition 4.14.** *Let $X$ be a Kummer surface. Suppose there is a lattice isomorphism $\phi : H^2(X;\mathbb{Z}) \to H^2(X;\mathbb{Z})$ satisfying*

*(1) $\phi$ preserves the classes of 16 exceptional rational curves and fixes one of them;*
*(2) $\phi|\Pi^\perp = id|\Pi^\perp$.*

*Then $\phi$ is the identity map.*

*Proof.* Because $\omega_X \in \Pi^\perp \otimes \mathbb{C}$, $\phi$ preserves the non-vanishing holomorphic 2-form. By lemma 4.11, $\phi$ induces an affine transformation on $I$. We use the relation between $\Pi$ and $\Pi^\perp$ that $\Pi^*/\Pi \cong (\Pi^\perp)^*/\Pi^\perp$ to show that the affine map on $I$ must be the identity. Therefore $\phi$ is trivial on a finite-index sublattice $\Pi \oplus \Pi^\perp$ of $H^2(X;\mathbb{Z})$ and is hence the identity. For details, see [Kon20, p.99]. □

Finally, in order to use the Torelli theorem for complex torus, we need to verify the second condition in theorem 4.2. There is a very natural map on $I$ obtained through lemma 4.11. Consider the situation in lemma 4.11. Let $\tau : I \to I'$ be the affine map induced by $\phi$ and assume it is linear by composing it with a translation. As $\phi$ maps the 16 exceptional curves to 16 exceptional curves, it maps $\Pi$ to $\Pi'$. Therefore it maps $\Pi^\perp$ to $\Pi'^\perp$. By proposition 4.9, $\phi$ induces an isomorphism $\psi : H^2(A;\mathbb{Z}) \to H^2(A';\mathbb{Z})$. Let $\tau^*$ be the dual map on cohomology group and $\psi_2 = \psi \bmod 2$. We have the following lemma.

**Lemma 4.15.** $\psi_2 = (\tau^*)^{-1} \wedge (\tau^*)^{-1}$.

*Proof.* This is mainly a verification. See [Kon20, p.101]. □

*Proof of Theorem 4.6.* By lemma 4.8, $X'$ is also a projective Kummer surface. Let $X = \mathrm{Km}(A)$, $X' = \mathrm{Km}(A')$. As $\phi$ maps the 16 exceptional curves to 16 exceptional curves, it maps $\Pi$ to $\Pi'$. Therefore it maps $\Pi^\perp$ to $\Pi'^\perp$. By proposition 4.9, $\phi$ induces an isomorphism $\psi : H^2(A;\mathbb{Z}) \to$

$H^2(A';\mathbb{Z})$. Because $\phi$ preserves the non-vanishing holomorphic 2-form, so does $\psi$. By lemma 4.15, $\psi_2$ can be written as a wedge product of a map on $H^1$. By proposition 4.3, $\psi$ satisfies the conditions in theorem 4.2. Therefore, we may apply theorem 4.2 that there exists a unique isomorphism of complex tori $\tilde{\phi} : A' \to A$ with $\tilde{\phi}^* = \psi$. As $\tilde{\phi}$ preserves the 16 fixed points, this gives an isomorphism $\varphi : X' \to X$ with $\varphi^*|\Pi^\perp = \phi|\Pi^\perp$. By proposition 4.14, when $\varphi^*$ coincides with $\phi$ on $\Pi^\perp$, they agree on the whole cohomology group. That is, $\varphi^* = \phi$. We have shown the existence of such an isomorphism of complex manifolds that induces the prescribed isomorphism on the second cohomology groups. Using proposition 4.14 again, it is easy to see that such $\varphi$ is in fact unique. □

## 5. Proof of the Torelli theorem for K3 surfaces

In this section we will give a proof of the Torelli theorem for K3 surfaces. As mentioned in the introduction, we will adopt a limiting argument, using isomorphisms of Kummer surfaces. The foundation is the deformation theory of complex manifolds, developed by Kodaira and Spencer.

**Definition 5.1.** Let $\mathcal{Y}, B$ be connected complex manifolds. A holomorphic map $\pi : \mathcal{Y} \to B$ is called a *complex analytic family* if the following conditions hold:

(1) $\pi$ is proper;
(2) The Jacobian matrix $J(\pi)$ of $\pi$ has rank equal to $\dim B$ at every point.

Therefore, $Y_t = \pi^{-1}(t)$ is a submanifold of $\mathcal{Y}$ for any $t \in B$. We also call $\mathcal{Y}$ a *deformation family* for $Y_t$. The space $B$ is sometimes called the *base space*.

**Definition 5.2.** A complex analytic family $\pi : \mathcal{Y} \to B$ is called *complete* at $t_0 \in B$ if for any deformation family

$$\pi' : \mathcal{Y}' \to B', \quad Y = \pi'^{-1}(s_0) = \pi^{-1}(t_0), \quad s_0 \in B',$$

there exists a neighborhood $B'' \subset B'$ of $s_0$ and a holomorphic map $f : B'' \to B$ with $f(s_0) = t_0$ such that the complex analytic family $\pi'$ is the pullback of $\pi$ by $f$.

In plain words, a complete deformation family is one that contains all possible 'directions' of deformations.

Let $\pi : \mathcal{Y} \to B$ be a complex analytic family and let $\Theta_{Y_t}$ be the sheaf of holomorphic vector fields over $Y_t$. Let $U$ be a neighborhood of $t \in B$ with local coordinates $t = (t_1, ...t_m)$ with $m = \dim B$. There exists an open covering $\{\mathcal{U}_i\}_{i\in I}$ of $\pi^{-1}(U)$ with local coordinates $\mathcal{U}_i = U_i \times U$, $(z_i, t) = (z_i^1, ..., z_i^n, t_1, ..., t_m)$ such that there are transition functions

$$z_i^l = f_{ij}^l(z_j, t)$$

for $l = 1, ..., n$ and where $f_{ij}^l$ are holomorphic functions on $\mathcal{U}_i \cap \mathcal{U}_j$. The transition functions vary holomorphically depending on $t \in U$. Consider

$$\theta_{ij}^\alpha = \sum_{l=1}^{n} \frac{\partial f_{ij}^l(z_j, t)}{\partial t_\alpha} \frac{\partial}{\partial z_i^l} \in H^0(U_i \cap U_j, \Theta_{Y_t})$$

for $\alpha = 1, ..., m$. It can be shown that $\theta_{ij}^\alpha$ does not depend on the choice of local coordinates. Moreover, it satisfies the cocycle condition on $U_i \cap U_j \cap U_k$, namely $\theta_{ij}^\alpha = \theta_{ik}^\alpha + \theta_{kj}^\alpha$. Therefore, it determines an element in $H^1(Y_t, \Theta_{Y_t})$ (using the Čech cohomology interpretation of sheaf cohomology).

**Definition 5.3.** Denote

$$\frac{\partial Y_t}{\partial t_\alpha} \in H^1(Y_t, \Theta_{Y_t})$$

the cohomology class determined by $\{\theta_{ij}^\alpha\}$. For

$$\frac{\partial}{\partial t} = \sum_{\alpha=1}^{m} a_\alpha \frac{\partial}{\partial t_\alpha} \in T_t(B),\ a_\alpha \in \mathbb{C},$$

define

$$\frac{\partial Y_t}{\partial t} = \sum_{\alpha=1}^{m} \frac{\partial Y_t}{\partial t_\alpha},$$

which is called the *infinitesimal deformation* of $Y_t$. The linear map

$$\rho_t : T_t(B) \to H^1(Y_t, \Theta_{Y_t})$$
$$\frac{\partial}{\partial t} \mapsto \frac{\partial Y_t}{\partial t}$$

is called the *Kodaira-Spencer map*.

We state two results on complex analytic families without proof (see [Kon20, p.81, p.83]).

**Theorem 5.4.** *The fibers of a complex analytic family are diffeomorphic.*

**Theorem 5.5.** *Let $Y$ be a compact complex manifold with $H^2(Y, \Theta_Y) = 0$. Then there exists a complete deformation family of $\pi : \mathcal{Y} \to B, Y = \pi^{-1}(t_0)$ of $Y$ such that the Kodaira-Spencer map is an isomorphism.*

**Proposition 5.6.** *Any deformation of a K3 surface is also a K3 surface.*

*Proof.* Consider a deformation family $\pi : \mathcal{X} \to \mathcal{B}, X = X_{t_0} = \pi^{-1}(t_0)$. By theorem 5.4, for any fiber $X_t$, $H^1(X_t; \mathbb{Z}) = 0$, and therefore $X_t$ has irregularity 0. Moreover, the Chern class is invariant under deformation. This is because as the fibers are diffeomorphic, for each $t \in \mathcal{B}$, the tangent bundle can be described as $(TM, J_t)$ where $TM$ is isomorphic to the tangent bundle of $X$ and $J_t$ is a complex structure that varies continuously. As the Chern class is equal to the pullback of certain cohomology class of the universal bundle and is invariant under homotopy of such map ([MS74, p.163]), the Chern class is invariant under deformation. Since $K_X = -c_1(X)$, each $X_t$ has trivial canonical bundle. This shows that each fiber $X_t$ is a K3 surface. □

**Proposition 5.7.** *Let $X$ be a K3 surface. Then $\Theta_X \cong \Omega^1_X$ as sheaves. In particular,*

$$dim\, H^0(X, \Theta_X) = dim\, H^2(X, \Theta_X) = 0,\ dim\, H^1(X, \Theta_X) = 20.$$

*Proof.* This is due to the fact that the existence of a non-vanishing holomorphic 2-form provides a natural isomorphism locally, namely contraction. Cover $X$ by local coordinates $(U_i, z_i)$. On each open subset $U_i$ consider the isomorphism

$$\Theta_X(U_i) \to \Omega^1_X(U_i)$$
$$\theta \mapsto i_\theta\, \omega_X$$

These local maps patch together nicely and define an isomorphism of sheaves: $\Theta_X \to \Omega^1_X$. The dimension of the sheaf cohomology groups with respect to $\Theta_X$ follows. □

**Corollary 5.8.** *Let $X$ be a K3 surface. Then $X$ has a complete analytic family, as its deformation family, whose Kodaira-Spencer map is an isomorphism. The base space of the complete analytic family has dimension 20.*

Based on the deformation theory, we introduce the so-called *period map* for a marked K3 surface. By theorem 3.9, the second cohomology group of any K3 surface is isomorphic to an abstract lattice. We fix such an abstract lattice $L$ once and for all. Let $\alpha : H^2(X; \mathbb{Z}) \to L$ be a lattice isomorphism. Let $\omega_X$ be the non-vanishing holomorphic 2-form and $\omega = (\alpha \otimes \mathbb{C})(\omega_X) \in L \otimes \mathbb{C}$. It satisfies the Riemann relation $\langle \omega, \omega \rangle = 0, \langle \omega, \overline{\omega} \rangle > 0$. Thus we consider

$$\Omega = \{\omega \in \mathbb{P}(L \otimes \mathbb{C}) \mid \langle \omega, \omega \rangle = 0, \langle \omega, \overline{\omega} \rangle > 0\},$$

called the *period domain* of K3 surfaces.

**Definition 5.9.** A *marked K3 surface* is a pair $(X, \alpha_X)$ where $X$ is a K3 surface and $\alpha_X$ is a lattice isomorphism $\alpha_X : H^2(X;\mathbb{Z}) \to L$. Let $\mathcal{M}$ be the set of isomorphism classes of marked K3 surfaces. Define a set-theortic map

$$\lambda : \mathcal{M} \to \Omega,$$
$$(X, \alpha_X) \mapsto (\alpha_X \otimes \mathbb{C})(\omega_X).$$

The map $\lambda$ is called the *period map* of marked K3 surfaces.

Next we consider the case of a deformation family.

**Definition 5.10.** For any K3 surface $X$, by corollary 5.8, consider a complete analytic family $\pi : \mathcal{X} \to B$ with $X = X_{t_0} = \pi^{-1}(t_0)$ and assume that $B$ is contractible. Then the higher push-forward of the constant sheaf $R^2\pi_*(\underline{\mathbb{Z}})$ is a constant sheaf. It is characterized by the property that on the stalk,

$$R^2\pi_*(\mathbb{Z})_p = H^2(\pi^{-1}(p), \mathbb{Z}).$$

Let $\alpha : R^2\pi_*(\underline{\mathbb{Z}}) \to \underline{L}$ be an isomorphism of sheaves where $\underline{L}$ is the constant sheaf over $B$ of the lattice $L$. By considering the induced map on the stalks, this gives each fiber $X_t$ a marking $\alpha_{X_t}$. The map

$$\lambda : B \to \Omega,$$
$$t \mapsto (\alpha_{X_t} \otimes \mathbb{C})(\omega_{X_t}).$$

is called the *period map* of a complex analytic family $\pi$.

**Theorem 5.11** (Local Torelli Theorem)**.** *The derivative*

$$d\lambda_{t_0} : T_{t_0}B \longrightarrow T_{\lambda(t_0)}\Omega$$

*is an isomorphism. Therefore, $\lambda$ is a local isomorphism.*

**Lemma 5.12.** *There is a natural isomorphism*

$$T_{\lambda(t_0)}\Omega \cong Hom(H^{2,0}(X), H^{1,1}(X)).$$

*Proof.* See [Kon20, p.92]. □

*Proof of theorem 5.11.* The key is that the following diagram commutes:

$$\begin{array}{ccc} T_{t_0}B & \xrightarrow{d\lambda_{t_0}} & T_{\lambda(t_0)}\Omega \\ {\scriptstyle \rho_{t_0}}\downarrow & \nearrow & \\ H^1(X_{t_0}, \Theta_{X_{t_0}}) & & \end{array}$$

where $\rho_{t_0}$ is the Kodaira-Spencer map and the slant arrow is the isomorphism in proposition 5.7.

This is done by a direct calculation. Take $\frac{\partial}{\partial t} \in T_{t_0}B$. Choose a holomorphic map $b : \Delta \to B$ with $b(0) = t_0$, $\left(\frac{db}{ds}\right)(0) = \frac{\partial}{\partial t}$.

(1) The horizontal map gives

$$(d\lambda_{t_0})\left(\frac{\partial}{\partial t}\right) = \left(\frac{d}{ds}\omega_{b(s)}\right)(0).$$

Take local coordinates $z = (z_1, z_2)$ on $X_{t_0}$. Take coordinates on $\mathcal{X}$:

$$w_1 = w_1(z,t), \qquad w_2 = w_2(z,t)$$

such that $w_1(z,t_0)=z_1$, $w_2(z,t_0)=z_2$. The holomorphic 2-form $\omega$ can be expressed as

$$\omega_t=\frac{1}{2}\sum_{i,j=1}^{2}\psi_{ij}(w,t)\,dw_i(z,t)\wedge dw_j(z,t)$$

where $\psi_{ij}$ are holomorphic functions.

Now we compute the derivative in local coordinates:

$$\begin{aligned}\left(\frac{\partial}{\partial t}\omega_t\right)(t_0)&=\frac{1}{2}\sum_{i,j=1}^{2}\frac{\partial\psi_{ij}}{\partial t}(w,t)\,dw_i(z,t)\wedge dw_j(z,t)\\&\quad+\psi_{ij}(w,t)\,d\left(\frac{\partial w_i}{\partial t}(z,t)\right)\wedge dw_j(z,t)\\&\quad+\psi_{ij}(w,t)\,dw_i(z,t)\wedge d\left(\frac{\partial w_j}{\partial t}(z,t)\right)\\&=\frac{1}{2}\sum_{i,j=1}^{2}\frac{\partial\psi_{ij}}{\partial t}(w,t)\,dz_i\wedge dz_j\\&\quad+\psi_{ij}(z,t_0)(\partial+\bar\partial)\left(\frac{\partial w_i}{\partial t}(z,t)\right)\Big|_{t=t_0}\wedge dz_j\\&\quad+\psi_{ij}(z,t_0)dz_i\wedge(\partial+\bar\partial)\left(\frac{\partial w_j}{\partial t}(z,t)\right)\Big|_{t=t_0}.\end{aligned}$$

The $(1,1)$-part is

$$\sum_{i,j=1}^{2}\psi_{ij}(z,t_0)\bar\partial\left(\frac{\partial w_i}{\partial t}(z,t_0)\right)\Big|_{t=t_0}\wedge dz_j.$$

(2) The image of the Kodaira-Spencer map is

$$\rho_t\left(\frac{\partial}{\partial t}\right)=\sum_{i=1}^{2}\bar\partial\left(\frac{\partial w_i}{\partial t}(z,t)\right)\Big|_{t=t_0}\frac{\partial}{\partial z_i}.$$

The image under the isomorphism

$$H^1(X_{t_0},\Theta_{X_{t_0}})\cong H^1(X_{t_0},\Omega^1_{X_{t_0}})\cong H^{1,1}(X_{t_0})$$

is given by the contraction:

$$i_\omega\left(\sum_{i=1}^{2}\bar\partial\left(\frac{\partial w_i}{\partial t}(z,t)\right)\Big|_{t=t_0}\frac{\partial}{\partial z_i}\right)=\sum_{i,j=1}^{2}\psi_{ij}(z,t_0)\bar\partial\left(\frac{\partial w_i}{\partial t}(z,t)\right)\Big|_{t=t_0}\wedge dz_j.$$

Therefore, the diagram commutes and the claim is verified. □

With the local isomorphism theorem established, we will prove a density result, which is the first step in applying the limit argument. Here we will only look at a very special type of K3 surfaces: marked singular Kummer surfaces, where singular means that the Néron–Severi lattice $S_X$ has the largest possible rank: $\operatorname{rank}(S_X)=20$. Note that these Kummer surfaces are projective. The density argument goes in two steps: first we identify the subset of $\Omega$ that is represented by a marked singular Kummer surface; then we show such a subset is dense in $\Omega$.

The following identification is crucial:

$$\begin{aligned}&\Omega\longleftrightarrow G_2^+(L)\\&\omega\longleftrightarrow\operatorname{span}\{\operatorname{Re}(\omega),\operatorname{Im}(\omega)\}\end{aligned}$$

where $G_2^+(L)$ is the set of oriented positive-definite rank 2 sublattices of $L \otimes \mathbb{R}$. This is a bijection because of the Riemann relation satisfied by $\omega$.

Starting from $\omega$, the Néron-Severi lattice and the transcendental lattice is

$$S_X = \{x \in L \mid \langle x, \omega \rangle = 0\}, \qquad T_X = S_X^{\perp}.$$

For a singular Kummer surface, $\operatorname{rank}(T_X) = 2$. Moreover, $T_X \otimes \mathbb{R} \cong \operatorname{Span}\{\operatorname{Re}(\omega), \operatorname{Im}(\omega)\}$. Recall that $\Pi^{\perp} \cong U(2)^{\oplus 3}$ is a sublattice of $L$. We have

$$\Pi \subseteq S_X, \qquad T_X \subseteq \Pi^{\perp}.$$

Therefore, for any $x \in T_X$, $\langle x, x \rangle \equiv 0 \pmod 4$. The converse is in fact also true.

**Theorem 5.13.** *Let $T$ be a positive-definite lattice of rank 2 satisfying*

$$\forall x \in T, \qquad \langle x, x \rangle \equiv 0 \pmod 4.$$

*Then there exists a Kummer surface $X$ with $T_X \cong T$.*

*Proof.* By the relation between the canonical bundle of a complex torus and its associated Kummer surface, it suffices to find a complex torus with transcendental lattice $T(\frac{1}{2})$, which is an even lattice.

By proposition 3.12, there is a primitive embedding of lattices

$$T\left(\frac{1}{2}\right) \hookrightarrow U^{\oplus 3}.$$

Let $\Gamma$ be a free abelian group of rank 4 and embeds $T(\frac{1}{2})$ primitively into $\wedge^2\Gamma^* \cong U^{\oplus 3}$. Pick an oriented orthogonal basis $\{x, y\}$ of $T(\frac{1}{2}) \otimes \mathbb{R}$ with $x^2 = y^2$. Consider

$$\omega = x + iy \in \wedge^2\Gamma^*_{\mathbb{C}}.$$

We have

$$\langle \omega, \omega \rangle = 0, \qquad \langle \omega, \bar{\omega} \rangle > 0.$$

Therefore $\mathbb{C}\omega \subseteq \wedge^2\Gamma^*_{\mathbb{C}}$ is a 1-dimensional isotropic subspace. Moreover, $\mathbb{C}\omega$ is in the image of the Plücker embedding $G(1,3) \hookrightarrow \mathbb{P}(\wedge^2\Gamma^*_{\mathbb{C}})$ and therefore $w$ is a pure tensor. We write $\omega = \eta_1 \wedge \eta_2$ where $\eta_1, \eta_2 \in \Gamma^*_{\mathbb{C}}$. The Riemann condition implies that

$$\omega \wedge \bar{\omega} = \eta_1 \wedge \eta_2 \wedge \bar{\eta}_1 \wedge \bar{\eta}_2 > 0.$$

Let $H$ be spanned by $\eta_1, \eta_2$ and $\bar{H}$ is the conjugate. Then this implies $H \cap \bar{H} = \{0\}$ and therefore $\Gamma^*_{\mathbb{C}} \cong H \oplus \bar{H}$.

Consider the complex torus $A = \mathbb{C}^2/\Gamma$ with $H_1(A; \mathbb{Z}) \cong \Gamma$. By the construction above, $T_A \otimes \mathbb{R}$ is spanned by $\operatorname{Re}(\eta_1 \wedge \eta_2)$ and $\operatorname{Im}(\eta_1 \wedge \eta_2)$, which are $x$ and $y$ in the integral cohomology group. Because the embedding is primitive, $T_A$ is spanned by $x$ and $y$. Therefore,

$$T_A \cong T\left(\frac{1}{2}\right)$$

as desired. □

Now it suffices to show the following:

**Proposition 5.14.** *The set of 2-dimensional positive-definite subspaces in $L \otimes \mathbb{R}$ generated by lattices of rank 2 whose elements have norms divisible by 4 is dense in $G_2^+(L)$.*

The idea is simply to perturb the orthogonal basis of the lattice so that the norm of the basis will become divisible by 4. We first prove this lemma.

**Lemma 5.15.** *Let $m, n$ be natural numbers and $M$ a lattice. Suppose that the set*

$$\mathcal{R} = \{\mathbb{R}e \in \mathbb{P}(M \otimes \mathbb{R}) \mid e \text{ primitive and } \langle e, e \rangle = m \pmod n\}$$

*is non-empty. Then it is dense in $\mathbb{P}(M \otimes \mathbb{R})$.*

*Proof.* Let $e_0$ be a primitive element in $M$ with $\langle e_0, e_0\rangle = m \pmod n$. We will show that for any non-empty open subset $V$ of $\mathbb{P}(M\otimes\mathbb{R})$, there exists an element in $\mathcal{R}$ contained in $V$. Let $\mathbb{R}e \in V$ be a line generated by a primitive element in $M$. If we can choose $e = \pm e_0$, then we are done. Suppose not. Consider the primitive sublattice $M' = M \cap (\mathbb{Q}e + \mathbb{Q}e_0)$ in $M$. Since $e$ is primitive, we can pick $f \in M'$ such that $\{e, f\}$ is a basis of $M'$. Then $e_0 = ae + bf$ with $a, b \in \mathbb{Z}$ and $a, b$ coprime. Hence, for any $N \in \mathbb{N}$, $e_N := e_0 + Nbe = (a + Nb)e + bf$ is primitive too. When $N$ is a multiple of $n$, $\langle e_N, e_N\rangle = \langle e_0, e_0\rangle = m \pmod n$. Because $V$ is open, by choosing a sufficiently big $N$, we have $\mathbb{R}e_N = \mathbb{R}(e + \frac{1}{Nb}e_0) \in V$ and hence $e_N \in V \cap \mathcal{R}$. We have shown that $\mathcal{R}$ is dense in $\mathbb{P}(M\otimes\mathbb{R})$. □

*Proof of proposition 5.14.* We show that for any $P_0 \in G_2^+(L)$, there exists a lattice $T$ of rank 2 with the norm of every element being divisible by 4 and $T\otimes\mathbb{R}$ is sufficiently close to $P_0$. Let $\{e_1^0, e_2^0\}$ be an orthogonal basis of $P_0$. The lattice $L$ contains an element of norm 4 and by lemma 5.15, we may find a primitive element $e_1 \in L$ with $\langle e_1, e_1\rangle = 4 \pmod 8$ and $\mathbb{R}e_1$ is sufficiently close to $\mathbb{R}e_1^0$. By proposition 3.12, we may embed $\langle e_1\rangle$ primitively in $L$. The orthogonal complement $M$ contains a primitive element with norm 64. Again by lemma 5.15, we may find a primitive element $e_2 \in M$ with $\langle e_2, e_2\rangle = 0 \pmod{64}$ and $\mathbb{R}e_2$ is sufficiently close to $\mathbb{R}e_2^0$. The elements $e_1, e_2$ are perpendicular and generate a positive-definitive subspace $P$ that is sufficiently close to $P_0$.

Next we show that $T = P \cap L$ satisfies the desired condition that the norm of each element is divisible by 4. For any $x \in T$, $\langle e_1, e_2\rangle x - \langle e_1, x\rangle e_1 \in T$ is perpendicular to $e_1$ and is a mutliple of $e_2$. Because $e_2$ is primitive, there exists $n \in \mathbb{Z}$ such that

$$\langle e_1, e_1\rangle x = \langle e_1, x\rangle e_1 + ne_2.$$

By taking the norm, we have

$$\langle e_1, e_2\rangle^2\langle x, x\rangle = \langle e_1, x\rangle^2\langle e_1, e_1\rangle + n^2\langle e_2, e_2\rangle.$$

By comparing the 2-power of each side, we conclude that $\langle x, x\rangle$ is divisible by 4. □

**Remark 5.16.** Note that it is not necessarily true that $T$ is the integral span of $e_1$ and $e_2$, even if $e_1$ and $e_2$ are both primitive. Nevertheless, the equation $\langle e_1, e_1\rangle x = \langle e_1, x\rangle e_1 + ne_2$ still holds. This is why we need to choose a basis $e_1, e_2$ whose norms are divisible by specific 2-powers.

We have shown the following.

**Theorem 5.17.** *The set of periods of marked (singular) Kummer surfaces is dense in* $\Omega$.

Now we can prove that every K3 surface is diffeomorphic.

*Proof of theorem 2.11.* By the local Torelli theorem 5.11 and theorem 5.17, every K3 surface can be deformed into a Kummer surface and is diffeomorphic to it (theorem 5.4). Every Kummer surface is diffeomorphic as every complex torus with the 16 marked points is diffeomorphic. The result follows. □

**Corollary 5.18.** *K3 surfaces are simply-connected.*

*Proof.* Applying the Lefschetz hyperplane theorem to a quartic surface shows that $\pi_1(X) = 0$. □

Now we start the proof of the Torelli theorem for K3 surfaces (Theorem 1.3). We first consider the complex deformation families for $X$ and $X'$:

$$\pi : \mathcal{X} \to B, \qquad \pi' : \mathcal{X}' \to B$$

where $\pi^{-1}(0) = X$, $\pi'^{-1}(0') = X'$. We assume that they're complete and the Kodaira-Spencer maps are isomorphisms.

Now we give a marking for each of the family $\mathcal{X}$ and $\mathcal{X}'$. Consider a sheaf isomorphism

$$\alpha : R^2\pi_*(\mathbb{Z}) \longrightarrow \underline{L}.$$

We take another marking $\alpha'$ on $\mathcal{X}'$:

$$\alpha' : R^2\pi'_*(\mathbb{Z}) \longrightarrow \underline{L}$$

such that on the stalk at 0, $(\alpha')^{-1} \circ \alpha|_0 = \phi$. The composition

$$\Phi := (\alpha')^{-1} \circ \alpha : R^2\pi_*(\mathbb{Z}) \longrightarrow R^2\pi'_*(\mathbb{Z})$$

induces (on the stalk) a lattice isomorphism of the second cohomology groups of each K3 surface in one family to one in another family. For each $t \in B$, these are denoted

$$\phi_t : H^2(X_t; \mathbb{Z}) \longrightarrow H^2(X'_t; \mathbb{Z}).$$

Note that $\phi_0 = \phi$. The period maps $\tau : B \to \Omega, \tau' : B' \to \Omega$ associated with the analytic families $\pi, \pi'$ are local isomorphisms. Therefore, by possibly shrinking $B$ and $B'$, we may assume that the two families are parameterized by the same base space $B$, and that $\tau'^{-1} \circ \tau$ is the identity. Hence $\phi_t$ takes $\omega_{X_t}$ to $\omega_{X'_t}$.

By theorem 5.17, there is a dense subset $K$ of $B$ whose fibers of the analytic family $\pi$ are Kummer surfaces. Then, provided that we can verify the assumptions in theorem 4.6, there exists a unique isomorphism of complex manifold

$$\varphi_{t_i} : X'_{t_i} \longrightarrow X_{t_i}$$

such that $\varphi_{t_i}^* = \phi_{t_i}$ for every $t_i \in K$. The first assumption is satisfied as discussed above. What is left is the second assumption. Now we need to show the following statements in order:

(1) If $t$ is sufficiently close to 0, then $\phi_t$ sends the Kähler cone of $X'_t$ to that of $X_t$.
(2) By choosing a sequence $\{t_i\}$ in $B$ that converges to 0, we may find a limit $\varphi$ of $\varphi_{t_i}$ that gives an isomorphism between $X$ and $X'$.
(3) Such isomorphism is unique.

**Step I** By the previous marking

$$\alpha : R^2\pi_*(\mathbb{Z}) \longrightarrow L,$$

we may view all the Kähler cone of $X_t$ in the same space $L \otimes \mathbb{R}$. Consider

$$\bigcup_{t\in B} \mathcal{D}(X_t) \subseteq \bigcup_{t\in B} P^+(X_t) \subseteq \bigcup_{t\in B} H^{1,1}(X_t, \mathbb{R}) \subseteq L \otimes \mathbb{R}.$$

We have the following result.

**Proposition 5.19.** *The union of Kähler cones*

$$\bigcup_{t\in B} \mathcal{D}(X_t) \subseteq \bigcup_{t\in B} H^{1,1}(X_t, \mathbb{R})$$

*is an open set.*

*Proof.* See [Kon20, p.110]. □

**Theorem 5.20.** *When $t$ is sufficiently close to 0, $\phi_t$ sends the Kähler cone of $X'_t$ to that of $X_t$.*

*Proof.* Because $\phi_t$ are lattice isomorphisms, there is a homeomorphism

$$\bigcup_{t\in B} H^{1,1}(X_t, \mathbb{R}) \longrightarrow \bigcup_{t\in B} H^{1,1}(X'_t, \mathbb{R}).$$

If $\phi_t$ maps an element of the Kähler cone to one of the Kähler cone, then $\phi_t$ maps the Kähler cone to the Kähler cone. Therefore, because the union of Kähler cones is open, the subset

$$\{t \in B \mid \phi_t \text{ sends Kähler cone to Kähler cone}\}$$

is open. The theorem follows. □

**Step 2** The goal is to show the following:

**Theorem 5.21.** *Suppose there is a dense subset $K$ of $B$ such that $\phi_t$ $(t \in K)$ is induced by a unique isomorphism*

$$\varphi_t : X'_t \longrightarrow X_t$$

*of complex manifolds. Then there exists an isomorphism $\varphi : X' \to X$ with $\varphi^* = \phi$.*

We start with the following lemma due to Bishop [Bis64] which gives us the limit object we are looking for.

**Lemma 5.22.** *Suppose that $\{t_i\}$ is a sequence of points in $K$ that converges to $0$ and $\Gamma_i \subseteq X_{t_i} \times X'_{t_i}$ is the graph of the isomorphism $\varphi_{t_i}$. Suppose that the volume of $\Gamma_i$ with respect to a Hermitian metric on $X_{t_i} \times X'_{t_i}$ is bounded. Then for a subsequence of $\{t_i\}$, $\{\Gamma_i\}$ converges to a 2-dimensional complex analytic subspace $\Gamma_0$ in $X \times X'$.*

We first show that this applies to our case of K3 surfaces.

**Proposition 5.23.** *The volumes of $\Gamma_i$, which are the graphs of isomorphisms between Kummer K3 surfaces, are bounded.*

*Proof.* Let $\kappa_t, \kappa'_t$ be Kähler forms on $X_t$ and $X'_t$ which vary continuously with $t$. Let $p_1 : X_t \times X'_t \to X_t$, $p_2 : X_t \times X'_t \to X'_t$ be the projections. Then the volume is given by

$$\mathrm{vol}(\Gamma_i) = \int_{\Gamma_i} \left(p_1^*(\kappa_t) + p_2^*(\kappa'_t)\right)^2.$$

The graph $\Gamma_i$ is also the image of

$$F := \varphi_{t_i} \times 1_{X'_{t_i}} : X'_{t_i} \longrightarrow X_{t_i} \times X'_{t_i}.$$

Pulling back using $F$, we have

$$\begin{aligned}
\mathrm{vol}(\Gamma_i) &= \int_{X'_{t_i}} \left(F^* p_1^*(\kappa_{t_i}) + F^* p_2^*(\kappa'_{t_i})\right)^2 \\
&= \int_{X'_{t_i}} \left(((p_1 F)^*(\kappa_{t_i}))^2 + (1_{X'_{t_i}})^*(\kappa'_{t_i})^2 + 2(p_1 F)^*(\kappa_{t_i}) \wedge (1_{X'_{t_i}})^*(\kappa'_{t_i})\right) \\
&= \int_{X_{t_i}} \kappa_{t_i}^2 + \int_{X'_{t_i}} (\kappa'_{t_i})^2 + 2\int_{X'_{t_i}} \varphi_{t_i}^*(\kappa_{t_i}) \wedge \kappa'_{t_i}.
\end{aligned}$$

The first two terms involves $\kappa_{t_i}$ and $\kappa'_{t_i}$ which vary continuously and are therefore bounded. For the third term, notice the map $\varphi_{t_i}$ is undefined outside $K$ so we cannot immediately conclude that it is continuous. Here we use the fact that the Kähler classes are closed. Therefore,

$$\int_{X'_{t_i}} \varphi_{t_i}^*(\kappa_{t_i}) \wedge \kappa'_{t_i} = \langle [\varphi_{t_i}(\kappa_{t_i})], [\kappa'_{t_i}] \rangle = \langle [\phi(\kappa_{t_i})], [\kappa'_{t_i}] \rangle$$

and is continuous and bounded around 0. □

Therefore by lemma 5.22, the limit of $\Gamma_i$, denoted as $\Gamma_0 \in X \times X'$, is a 2-dimensional complex analytic manifold.

**Lemma 5.24.** *The set $\Gamma_0$ can be expressed as*

$$\Gamma_0 = \Delta_0 + \sum_{ij} a_{ij} C_i \times C'_j, \qquad a_{ij} \in \mathbb{N}$$

*where $\Delta_0$ is the graph of an isomorphism between $X$ and $X'$, and $C_i$ and $C'_j$ are irreducible curves on $X$ and $X'$ respectively.*

*Proof.* Let $p_1 : X \times X' \to X$, $p_2 : X \times X' \to X'$ be the projections. We first establish a relation of $\phi$ and $p_1, p_2$ and $[\Gamma_0] \in H^4(X \times X'; \mathbb{Z})$.

We describe a map induced from $[\Gamma_0]$:

$$[\Gamma_0]^* : H^i(X; \mathbb{Z}) \longrightarrow H^i(X'; \mathbb{Z}).$$

Start with $x \in H^i(X; \mathbb{Z})$, the map factors as

$$H^i(X; \mathbb{Z}) \longrightarrow H^{i+4}(X \times X'; \mathbb{Z}) \longrightarrow H^i(X'; \mathbb{Z})$$

$$x \longmapsto p_1^*(x) \smile [\Gamma_0] \longmapsto p_{2*}\left(p_1^*(x) \smile [\Gamma_0]\right)$$

where $p_{2*}$ is the Gysin map.

An equivalent definition is to take $x$ as a cocycle such that $\widetilde{x} \times X'$ is in general position with $\Gamma_0$ and then consider the projection of $(\widetilde{x} \times X') \cap \Gamma_0$ to $X'$ which defines a cocycle on $X'$.

We have $[\Gamma_0]^* = \phi$.

Now for each irreducible component $Z$ of $\Gamma_0$, we consider the image of the projection under $p_1$ and $p_2$. There are four cases:

(1) $p_1(Z)$ and $p_2(Z)$ are both 2-dimensional.
(2) $p_1(Z)$ is a curve and $p_2(Z)$ is 2-dimensional, or vice versa.
(3) $p_1(Z)$ is a point and $p_2(Z)$ is 2-dimensional, or vice versa.
(4) $p_1(Z)$ and $p_2(Z)$ are both curves.

The other cases do not appear as then $Z$ cannot be 2-dimensional.

For $\Gamma_i$ and the related projection maps, $p_1$ and $p_2$ have degree 1. Therefore, the limit also has both projection map degree 1. In the case of (2),(3),(4), one of $p_1$ and $p_2$ factors through a point or a curve. But

$$\phi(x) = p_{2*}(\langle p_1^*(x), [\Gamma_0] \rangle)$$

sends $H^{2,0}(X)$ to $H^{2,0}(X')$. Therefore, one and only one irreducible component of type (1) exists. Because degree is additive, type (2) and (3) cannot exist. Thus, $\Gamma_0$ is the sum of an irreducible component of type (1) and finitely many product of irreducible curves.

Finally, denote the irreducible component of type (1) by $\Delta_0$. We have that

$$p_1 : \Delta_0 \to X, \qquad p_2 : \Delta_0 \to X'$$

both having degree 1. Therefore $\Delta_0$ and $X$, $\Delta_0$ and $X'$ are bimeromorphic. Thus $X$ and $X'$ are bimeromorphic. Any two bimeromorphic complex surfaces are related by a finite sequence of blow ups and blow downs. But K3 surfaces are minimal (see proposition 2.9). Therefore, $p_2^{-1} \circ p_1$ gives a biholomorphism between $X$ and $X'$. This finishes the proof. Notice that the map $(p_2^{-1} \circ p_1)^*$ on $H^2(X; \mathbb{Z})$ coincides with $\phi$ by our construction of $[\Gamma_0]^*$ above. □

**Lemma 5.25.** *In the previous lemma, $a_{ij} = 0$ for any $i, j$.*

*Proof.* This a clever computation using a Kähler class $\kappa$.

By the second definition of $[\Gamma_0]^*$, we have

$$\phi(\kappa) = [\Gamma_0]^*(\kappa) = \kappa + \sum_{ij} a_{ij} \langle \kappa, C_i \rangle C_j'.$$

Now consider

$$\langle \phi(\kappa), \phi(\kappa) + \kappa \rangle = \langle \kappa, \phi(\kappa) + \kappa \rangle + \sum_{ij} a_{ij} \langle \kappa, C_i \rangle \langle C_j', \kappa + \phi(\kappa) \rangle.$$

Thus we have

$$\begin{aligned}\sum_{ij} a_{ij}\langle\kappa, C_i\rangle\langle\phi(\kappa)+\kappa, C_j'\rangle &= \langle\phi(\kappa)+\kappa, \phi(\kappa)-\kappa\rangle \\ &= \langle\phi(\kappa),\phi(\kappa)\rangle - \langle\kappa,\kappa\rangle \\ &= 0.\end{aligned}$$

The last equality is due to the fact that $\phi$ is a lattice isomorphism. Because $\kappa$ and $\phi(\kappa)+\kappa$ are Kähler classes,

$$\langle\kappa, C_i\rangle > 0, \qquad \langle\phi(\kappa)+\kappa, C_j'\rangle > 0$$

for any irreducible curves $C_i$ and $C_j'$. Therefore, $a_{ij}=0$ for any $i,j$. □

**Step 3** This part is about the uniqueness of $\varphi$. It is equivalent to show the following:

**Theorem 5.26.** *Any automorphism of a K3 surface $X$ that acts trivially on $H^2(X;\mathbb{Z})$ is the identity.*

*Proof.* The automorphism group $\mathrm{Aut}(X)$ is a Lie group. [Kob95, p.77] Moreover, it has Lie algebra

$$T_{\mathrm{Id}}\,\mathrm{Aut}(X) \cong H^0(X,\Theta_X)$$

where $\Theta_X$ is the sheaf of holomorphic vector field. By proposition 5.7, for a K3 surface $X$, $H^0(X,\Theta_X)$ is trivial. This means that $\mathrm{Aut}(X)$ as a Lie group must be discrete.

By theorem 5.21, $\mathrm{Aut}(X)$ is sequentially compact and thus compact. Therefore $\mathrm{Aut}(X)$ is finite. Every element in $\mathrm{Aut}(X)$ has finite order.

Now take $g\in\mathrm{Aut}(X)$, $g\neq\mathrm{Id}$. Let $g$ have order $n$. If $p\in X$ is a fixed point of $g$, locally around the point $p$, $g$ can be realized as a linear function. Because $g$ is not the identity, the action of $dg$ on $T_pX$ is not trivial. Because $g$ preserves the non-vanishing holomorphic 2-form, $dg$ has determinant 1. Therefore, $dg$ is in the form

$$\begin{pmatrix}\epsilon & 0\\ 0 & \epsilon^{-1}\end{pmatrix}$$

where $\epsilon$ is a primitive $n$th root of unity. Therefore, the fixed points of $g$, if exists, are isolated. Because $X$ is compact, the number of fixed points is finite. By the Lefschetz fixed point theorem, the number of fixed points is equal to the Euler characteristic, which is 24.

Now consider the space

$$Y = X/\langle g\rangle.$$

It has 24 rational double points of type $A_{n-1}$ (see [Kon20, p.63]) at the fixed points of $g$. Let $f:\widetilde{Y}\to Y$ be the minimal resolution of $Y$, which is the blow-up at the 24 points. The non-vanishing holomorphic 2-form lifts to $\widetilde{Y}$. However, $H^2(\widetilde{Y};\mathbb{Z})$ has rank at least 24 for each exceptional divisor at the fixed points. By the classification of compact complex manifolds, this is impossible. □

*Proof of the Torelli theorem 1.3.* By theorem 5.20, $\phi_t$ preserves the Kähler cone and the non-vanishing holomorphic 2-form in a small neighborhood around 0. By theorem 5.17, in this small neighborhood, there is a dense set $K$ where the fibers are Kummer K3 surfaces. By theorem 4.6, there is $\varphi_t: X_t'\to X_t$ isomorphisms of complex manifolds such that $\phi_t=\varphi_t^*$ for every $t\in K$. By theorem 5.21, the limit of the graphs gives an isomorphism $\varphi: X'\to X$ of complex manifold with $\varphi^*=\phi$. Such isomorphism is unique by theorem 5.26. □

## 6. Surjectivity of the period map

In this section, we define a marked K3 surface as a triple $(X, \alpha, \kappa)$ where $\alpha : H^2(X, \mathbb{Z}) \to L$ a lattice isomorphism and $\kappa$ a Kähler class. Denote $\widetilde{\mathcal{M}}$ as the set of isomorphism class of marked K3 surfaces. There is a period map

$$\tilde{\lambda} : \widetilde{\mathcal{M}} \longrightarrow \widetilde{\Omega}^\circ$$
$$(X, \alpha, \kappa) \longmapsto (\alpha(\omega_X), \alpha(\kappa))$$

where

$$\widetilde{\Omega}^\circ := \{(\omega, \kappa) \in \Omega \times (L \otimes \mathbb{R}) \mid \langle \omega, \omega \rangle = 0,\ \langle \omega, \bar{\omega} \rangle > 0;\ \langle r, \kappa \rangle \neq 0,\ \forall r \in \Delta_\omega\}$$

where

$$\Delta_\omega = \{\delta \in L \mid \delta^2 = -2,\ \langle \delta, \omega \rangle = 0\}.$$

The Torelli theorem 1.3 states that $\tilde{\lambda}$ is injective. In this section we prove the following result.

**Theorem 6.1.** *The map $\tilde{\lambda}$ is surjective.*

An important ingredient is the following construction. For $(\omega, \kappa) \in \widetilde{\Omega}^\circ$, consider the 3-dimensional space $E(\omega, \kappa) \subseteq L \otimes \mathbb{R}$ generated by $\mathrm{Re}(\omega), \mathrm{Im}(\omega), \kappa$. This is positive-definite and $\{\mathrm{Re}(\omega), \mathrm{Im}(\omega), \kappa\}$ is an orthogonal basis. This gives a map

$$\pi : \widetilde{\Omega}^\circ \longrightarrow G_3^+(L \otimes \mathbb{R})$$
$$(\omega, \kappa) \longmapsto E(\omega, \kappa)$$

where $G_3^+(L \otimes \mathbb{R})$ denotes the set of 3-dimensional oriented subspaces in $L \otimes \mathbb{R}$.

**Theorem 6.2.** *The image of $\tilde{\lambda} : \widetilde{\mathcal{M}} \to \widetilde{\Omega}^\circ$ consists of fibers of $\pi$.*

To prove this result, we need to take a digression on hyperkähler structure and some results from differential geometry. The reference for this part is [Bar+15].

Let $X$ be a smooth manifold. An *almost-quaternionic structure* on $X$ consists of two almost-complex structures $I, J$ such that $IJ + JI = 0$. Note that if we let $K = IJ$, then $I^2 = J^2 = K^2 = IJK = -1$ and indeed $a + bI + cJ + dK$ $(a, b, c, d \in \mathbb{R})$ forms the quaternion algebra. For every $Q = a + bI + cJ + dK$, $Q^2 = -1$ if and only if $b^2 + c^2 + d^2 = 1$. Thus for every almost-quaternionic structure $\{I, J\}$, there is associated a *sphere of almost-complex structures*

$$S(I, J) := \{bI + cJ + dK \mid b^2 + c^2 + d^2 = 1\}.$$

For $(X, g)$ a Riemannian manifold, an almost-quaternionic structure $\{I, J\}$ is called a *quaternionic structure* if $I, J$ are Kähler with respect to $g$, meaning $I, J$ are parallel to the Levi-Civita connection $\nabla$. Then any element in $S(I, J)$ is also Kähler with respect to $g$ and in this case $S(I, J)$ is called *the sphere of complex structures* associated to $\{I, J\}$.

**Lemma 6.3.** *Let $X$ be a simply-connected complex manifold with a metric $g$ that is Kähler with respect to the underlying complex structure $I$. Then,*

1. *If $\mathrm{Ric}(g) = 0$, the canonical bundle is trivialized by a flat section $\omega_I$.*
2. *If moreover $X$ is a complex surface, there is an almost-complex structure $J$ such that $\{g, I, J\}$ is a quaternionic structure on $X$. It is obtained as follows: define $J_1 : T_pX \to T_pX$ such that $g(J_1 \cdot, \cdot) = \mathrm{Re}\, \omega_I(\cdot, \cdot)$. Then $J = \lambda J_1$ for a suitable $\lambda \in \mathbb{C}^*$ is as desired.*

*Proof.* See [Bar+15, p.336]. □

To apply this to K3 surfaces, we need the following deep result due to Yau.

**Theorem 6.4** ([Yau77])**.** *Let $X$ be a compact complex manifold with trivial canonical bundle and of dimension at least two. If an element of $H^2(X; \mathbb{R})$ can be represented by a Kähler form, then it can be represented by a unique Kähler form $w_g$ with $\mathrm{Ric}(g) = 0$.*

Applying this theorem, we obtain the following result.

**Theorem 6.5.** *Let $X$ be a K3 surface and $\kappa$ a Kähler class. Then there exists a unique Ricci-flat metric $g$ representing $\kappa$ and another complex structure $J$ on $X$ such that $\{g, I, J\}$ is a quaternionic structure.*

While $g$ is unique, the complex structure $J$ is not unique. Nevertheless, the following holds:

**Proposition 6.6.** *Let $J'$ be another complex structure that satisfies the conditions of theorem 6.5. Then $J' \in S(I, J)$.*

*Proof.* We first pick any point $p \in X$ and show that at this point, $J'_p = aJ_p + bK_p$. We choose any non-zero vector $e \in T_pX$. Then $\{e, Ie, Je, Ke\}$ is an orthogonal basis. Because $g$ is also Kähler with respect to the complex structure $J'$, we have $\langle J'u, v\rangle = \langle J'^2u, J'v\rangle = -\langle u, J'v\rangle$. Therefore,

$$\langle J'e, e\rangle = -\langle e, J'e\rangle,$$
$$\langle J'e, Ie\rangle = -\langle e, J'Ie\rangle = -\langle Ie, IJ'Ie\rangle = -\langle J'e, Ie\rangle.$$

Thus $J'e = aJe + bKe$. Next we have

$$J'(Ie) = I(J'e) = I(aJe + bKe) = bJe - aKe.$$

Similarly, we can compute

$$\langle J'(Je), e\rangle = -\langle Je, J'e\rangle = -a,$$
$$\langle J'(Je), Ie\rangle = -\langle Je, J'(Ie)\rangle = -b,$$
$$\langle J'(Je), Je\rangle = -\langle Je, J'(Je)\rangle.$$

So $J'(Je) = -ae - bIe$. We also compute that $J'(Ke) = -be + aIe$. Comparing the coefficients shows that $J' = aJ + bK$. We have shown that at $T_pX$, $J'_p = aJ_p + bK_p$. Next we write $J(p) = a(p)J + b(p)K$. That $J'$ is parallel to the Levi-Civita connection shows that

$$\begin{aligned} 0 = \nabla J' &= \nabla(cJ + dK) \\ &= (da)J + (db)K \end{aligned}$$

because that $I, J, K$ are all parallel to the Levi-Civita connection. Therefore $a(p), b(p)$ are constant for points $p \in X$. We have shown $J' \in S(I, J)$. □

Therefore there is a unique sphere associated to $(X, \kappa)$ and we call it the *canonical sphere of complex structures*, denoted by $S(X)$.

For any tangent vector $Y \in T_pX$, $\{Y, IY, JY, KY\}$ is an orthogonal basis. Let $\{\varphi, \varphi_I, \varphi_J, \varphi_K\}$ be the dual basis of $T_p^*X$. For each $H \in S(X)$, associate to it a Kähler form

$$\kappa_H(Y, Z) = g(Y, HZ).$$

Now we have that at the point $p$:

$$\kappa_J = \mu \operatorname{Re} \omega_I, \qquad \kappa_K = \mu \operatorname{Im} \omega_I.$$

Therefore,

$$E(\omega, \kappa) = \operatorname{span}\{\kappa_I, \kappa_J, \kappa_K\}.$$

Now take a K3 surface with a fixed Kähler class $(X, \kappa)$. Its complex structure is denoted $I$. For any $\kappa_0 \in E(\omega, \kappa)$,

$$\kappa_0 = b\kappa_I + c\kappa_J + d\kappa_K.$$

Rescale it such that

$$\mu\kappa_0 = \mu(b\kappa_I + c\kappa_J + d\kappa_K)$$

and $\mu^2(b^2+c^2+d^2)=1$. Then $\mu\kappa_0=\kappa_H$ for some complex structure $H\in S(X)$. Then notice that the orthogonal complement of $\kappa_H$ in $E$ is precisely $E(\omega_H)$. Any 3-space in $L\otimes\mathbb{R}$ that comes from $E(\omega,\kappa)$ must be in such form. Finally, the assignment

$$H\longmapsto E(\omega_H)$$

for $H\in S(X)$ is surjective. This is because $H=bI+cJ+dK$ subject to $b^2+c^2+d^2=1$ but has no restriction on the direction. Therefore, every placement of $E(\omega)$ in $E$ comes from a choice of complex structure in $S(X)$ on the K3 surface $(X,\kappa)$. We have proved theorem 6.2.

After this result, the proof goes as follows.

**Lemma 6.7.** *Let $(\omega,\kappa)\in\widetilde{\Omega}^\circ$. If $E(\omega,\kappa)\cap L$ contains a primitive sublattice $M$ of rank 2 satisfying*

$$\langle x,x\rangle\equiv 0 \pmod 4,\qquad \forall x\in M$$

*then $(\omega,\kappa)\in\mathrm{Im}(\tilde{\lambda})$.*

*Proof.* By theorem 6.2, we may assume $M\subseteq E(\omega)$. Then there exists a marked K3 surface (Kummer surface) $(X,\alpha_X)$ with $\alpha_X(\omega_X)=\omega$. By possibly composing a reflection to $\alpha_X$, we may assume $\alpha_X^{-1}(\kappa)\in\mathcal{D}(X)$. Now every element in the Kähler cone is a Kähler class. Therefore

$$\tilde{\lambda}(X,\alpha_X,\alpha_X^{-1}(\kappa))=(\omega,\kappa).$$

We have $(\omega,\kappa)\in\mathrm{Im}(\tilde{\lambda})$. □

**Lemma 6.8.** *Let $(\omega,\kappa)\in\widetilde{\Omega}^\circ$. If $E(\omega,\kappa)\cap L$ contains a primitive element $x$ such that*

$$\langle x,x\rangle\equiv 0 \pmod 4,$$

*then $(\omega,\kappa)\in\mathrm{Im}(\tilde{\lambda})$.*

*Proof.* By theorem 6.2, we may assume $x\in E(\omega)$. By lemma 5.15, there is a dense subset of $L\otimes\mathbb{R}$ whose elements $y\in L$ satisfy $\langle y,y\rangle\equiv 0 \pmod 4$ and are primitive. If $y$ is not a multiple of $x$, $M=\mathbb{Z}x+\mathbb{Z}y$ satisfies the condition above. If $y\in E(\omega)$, then we are done. If $y\notin E(\omega)$, choose $\eta\in\mathcal{D}(x)$ such that $M\subseteq E(\omega,\eta)$. This is always possible because $\mathcal{D}(x)$ is a convex cone. Such $\eta$ exists densely in $\mathcal{D}(x)$. Then there is a marked K3 surface $(X,\alpha_X,\eta)$ such that

$$\tilde{\lambda}(X,\alpha_X,\eta)=(\omega,\eta).$$

Finally we may switch a Kähler class in $X$ to be $\kappa$. □

*Proof of theorem 6.1.* Let $(\omega,\kappa)$ be any element in $\widetilde{\Omega}^\circ$. If there exists an element that is primitive and whose norm is divisible by 4 in $E(\omega)$, then $(\omega,\kappa)\in\mathrm{Im}(\tilde{\lambda})$. If not, there is a dense subset of such element $x$ in $E(\omega)\times\mathcal{D}(x)$. Let $\eta\in\mathcal{D}(x)$ such that $x\in E(\omega,\eta)$. Then $(\omega,\eta)\in\mathrm{Im}(\tilde{\lambda})$. Again now we may simply choose another Kähler class $\kappa$ in the Kähler cone. □

With this result, we finally prove theorem 2.12 that not all K3 surfaces are projective.

*Proof of theorem 2.12.* Because the map $\tilde{\lambda}$ is surjective, there exists a marked K3 surface such that $\alpha_X(\omega_X)\in L\otimes\mathbb{C}$ is an irrational point. In such case, the Néron–Severi lattice $S_X$ is empty as any integer point will not lie in the orthogonal complement of $\alpha_X(\omega_X)$. By proposition 3.21, such K3 surface is not projective. □

## Acknowledgments

I would like to express my sincere gratitude to my mentor Dr. Sidhanth Raman for his patience and unwavering support. This project would not have been possible without his guidance. I would also like to thank Prof. Peter May for his dedicated efforts in organizing the REU. It has been a fascinating summer without parallel.

## References


[Bar+15] Wolf P. Barth et al. *Compact Complex Surfaces*. Springer Berlin, Heidelberg, 2015. ISBN: 978-3-642-57739-0. URL: https://link.springer.com/book/10.1007/978-3-642-57739-0.

[Bis64] Errett Bishop. "Conditions for the analyticity of certain sets". In: *Michigan Math. J.* 11 (1964), pp. 133–153.

[Har77] Robin Hartshorne. *Algebraic Geometry*. Springer New York, NY, 1977. ISBN: 978-0-387-90244-9. URL: https://link.springer.com/book/10.1007/978-1-4757-3849-0.

[Hat22] Allen Hatcher. *Algebraic Topology*. 2022. URL: https://pi.math.cornell.edu/~hatcher/AT/ATpage.html.

[Kob95] Shoshichi Kobayashi. *Transformation Groups in Differential Geometry*. Springer Berlin, Heidelberg, 1995. ISBN: 978-3-540-58659-3.

[Kon20] Shigeyuki Kondō. *K3 Surfaces*. European Mathematical Society Press, 2020. ISBN: 978-3-03719-708-0. URL: https://ems.press/books/etm/181.

[LP80] Eduard Looijenga and Chris Peters. "Torelli theorems for Kähler K3 surfaces." In: *Compositio Mathematica* 42 (1980), pp. 145–186. URL: https://www.numdam.org/item/CM_1980__42_2_145_0/.

[MS74] John W. Milnor and James D. Stasheff. *Characteristic Classes. (AM-76)*. Princeton University Press, 1974. ISBN: 9780691081229. URL: http://www.jstor.org/stable/j.ctt1b7x751.

[Siu83] Y.T. Siu. "Every K3 surface is Kähler". In: *Invent Math* 73 (1983), pp. 139–150.

[Yau77] Shing-Tung Yau. "Calabi's conjecture and some new results in algebraic geometry". In: *Proceedings of the National Academy of Sciences* 74.5 (1977), pp. 1798–1799. DOI: 10.1073/pnas.74.5.1798. eprint: https://www.pnas.org/doi/pdf/10.1073/pnas.74.5.1798. URL: https://www.pnas.org/doi/abs/10.1073/pnas.74.5.1798.